

DECLARED GEOMETRY: A GEOMETRIC PRIOR BOTTLENECK

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ABSTRACT

Representations that generalize must discard target-irrelevant input variation while preserving target-relevant information in a predictive geometry. Geometric approaches make target geometry explicit but leave compression implicit; information-bottleneck (IB) methods make this compression explicit but do not condition it on the target, a gap the conditional entropy bottleneck (CEB) closes with a label-conditional reference. Yet a further limitation remains: its per-label KL imposes no cross-label constraint, and task information collapses as the residual approaches zero, so CEB cannot by itself select a stable, task-preserving path. To address this, we introduce the geometric prior bottleneck (GPB) framework, which places a declared geometry directly in the label-conditional reference: OPB predicts classification from an equal-radius orthogonal frame through anchor energies, and EPB predicts regression from an isometric label axis through a projection tied to the same axis. GPB preserves the CEB certificate and yields comparator-dependent alignment and separation guarantees that combine into an explicit β -dependent geometry-transfer bound with a comparator-specific floor. Across twelve dataset–backbone settings with complete IB and non-IB baselines, plus controlled, geometric, compression, robustness, and ablation studies, GPB is best or tied-best in eight settings and achieves a higher mean than CEB in all twelve.

1 INTRODUCTION

Generalization requires a balance between compression and structure: a representation should discard input variation irrelevant to the target while preserving target-relevant information in a geometry that supports reliable prediction.

Geometric approaches focus on how target information is organized within a representation. Across classification and regression, neural-collapse-inspired frames, equidistant or orthogonal constructions, fixed classifiers, and isometric embeddings or readouts impose structure on features, prototypes, model weights, or prediction axes (Papayan et al., 2020; Deng et al., 2014; Xu et al., 2022; Tong & Davis, 2023; Koromilas et al., 2026; Pai et al., 2022; Luo et al., 2023; Belucci et al., 2026). These methods make target geometry explicit, but their objectives primarily shape the structure needed for prediction. They do not generally include a separate criterion for controlling the target-irrelevant input variation retained in the representation, leaving compression implicit.

Information-bottleneck methods such as VIB, NIB, and DVCCA make compression explicit by limiting the information a representation retains from the input through marginal-prior matching, prior-free estimation, or reconstruction-augmented objectives (Alemi et al., 2017; Kolchinsky et al., 2019; Abdelaleem et al., 2025). Despite their different formulations, their compression mechanisms are not conditioned on the target label and therefore do not isolate the input variation retained beyond it. CEB addresses this limitation by introducing a label-conditional reference and directly targeting the residual input information that remains after the label is known (Fischer, 2020).

A certificate-plane diagnostic (Fig. 3d) exposes a further limitation of CEB under strong compression: task information, retained along most of the trajectory, collapses sharply as the estimated residual information approaches zero. This is not evidence that residual compression itself destroys task information; CEB does not by itself select a stable, task-preserving compression path. The cause is the freedom of its label-conditional reference to adapt to the posterior: the KL is computed

label by label, with no constraint on the pairwise relations among label conditionals, so the reference absorbs the compression pressure by shrinking or reorganizing its conditionals and loses the cross-label geometry that could have served as a stable target.

To address this, we introduce the geometric prior bottleneck (GPB) framework, which places a declared geometry directly in the label-conditional reference. A GPB prior is a conditional Gaussian with fixed unit isotropic covariance whose means satisfy a prescribed separation structure for discrete labels or a prescribed label metric for continuous targets. The form and scale of this geometry are chosen before training, while its overall orientation can remain learnable. We instantiate GPB in two forms. For classification, the orthogonal prior bottleneck (OPB) constructs an equal-radius orthogonal frame and predicts using the anchor energies induced by that frame. For regression, the equidistant prior bottleneck (EPB) maps labels onto an isometric axis and predicts through a projection tied to the same axis. In both cases, the same prior means serve as KL targets and prediction anchors, unifying information compression with explicit target geometry.

Our analysis connects the declared prior geometry to the learned posterior. Theorem 2.1 bounds the average anchor mismatch between posterior means and their prescribed anchors by a comparator-dependent certificate. Theorem 2.2 turns measured mismatch into a lower bound on the distance between posterior centers of different classes. Corollary 2.3 combines the two into the main end-to-end guarantee: with zero comparator KL floor, the certified deficit from the declared separation is $O(\beta^{-1/2})$, while a non-zero comparator KL imposes a comparator-specific floor. The guarantees certify geometry transfer rather than superiority over unconstrained references.

We evaluate GPB across twelve dataset–backbone settings spanning classification and regression. GPB is best or tied-best in eight settings and achieves a higher mean than CEB in all twelve. Controlled studies examine the comparator-dependent floor, while geometric diagnostics trace β -dependent mismatch and transfer of the declared structure to posterior class centers. Compression sweeps and targeted robustness experiments reveal favorable behavior in regimes where CEB degrades, and factor-wise ablations distinguish the contribution of the geometric bottleneck from that of the readout.

In summary, our contributions are: (i) identifying and formalizing the missing cross-label geometric constraint in CEB; (ii) introducing the GPB framework, with OPB and EPB, to unify conditional compression with explicit target geometry; (iii) establishing an upper bound on average anchor mismatch and a lower bound on posterior class-center separation, together with an explicit β -dependent geometry-transfer rate; and (iv) demonstrating the method across classification and regression through benchmark comparisons, geometric diagnostics, compression studies, robustness experiments, and factor-wise ablations.

2 METHOD

2.1 THE GEOMETRIC PRIOR BOTTLENECK FRAMEWORK

Let $Y - X - Z$ and let the encoder posterior be $q_\phi(z|x) = \mathcal{N}(\mu_\phi(x), \text{diag } \sigma_\phi^2(x))$. CEB (Fischer, 2020) replaces the VIB marginal reference by a label-conditional reference $r(z|y)$. For any choice of this reference, the certificate

$$I(X; Z|Y) \leq \mathbb{E} \text{KL}(q_\phi(z|X) \| r(z|Y)), \quad (1)$$

upper-bounds the residual input information $I(X; Z|Y)$, which CEB minimizes as its compression criterion. The certificate is an average of within-sample terms, each involving only the reference of the sample’s own label: no term couples two distinct label conditionals. Consequently, it records neither the relative directions nor the pairwise distances of the class-conditional means, so the compression criterion alone cannot prefer any cross-label geometry.

The GPB framework. GPB is not a single prior but a framework for declaring reference geometry: it restricts the reference to conditional Gaussians

$$p_\theta(z|y) = \mathcal{N}(m_\theta(y), \tau^2 I), \quad \tau^2 = 1, \quad (2)$$

where p_θ is the label-conditional Gaussian reference, $m_\theta(y)$ its anchor mean, and the prior covariance is fixed and isotropic. The posterior covariance remains the learned diagonal $\text{diag } \sigma_\phi^2(x)$. Their

means obey a declared two-sided separation for every pair of distinct discrete labels $y \neq y'$,

$$\Delta \leq \|m_\theta(y) - m_\theta(y')\| \leq \bar{\Delta}, \quad (3)$$

where Δ and $\bar{\Delta}$ are the declared minimal and maximal anchor distances (the separation margin and its ceiling), or a two-sided label metric condition for P_Y -almost every label pair (y, y') , where d_Y is a metric on the label space $\mathcal{Y}$:

$$\rho_{\min} d_Y(y, y') \leq \|m_\theta(y) - m_\theta(y')\| \leq \rho_{\max} d_Y(y, y'), \quad (4)$$

so the anchor map $y \mapsto m_\theta(y)$ is a bi-Lipschitz embedding of the label metric, with lower and upper constants $\rho_{\min}$ and $\rho_{\max}$. All constants are fixed before training. These conditions define the geometric prior family $\mathcal{G}$. The objective is

$$\mathcal{L}_{\text{GPB}} = \mathbb{E}[\ell(c_\psi(z), Y) + \beta \text{KL}(q_\phi(z|X) \| p_\theta(z|Y))], \quad z \sim q_\phi(z|X). \quad (5)$$

The instances below, OPB (classification) and EPB (regression), realize this single objective with $\ell \geq 0$ cross-entropy for classification and the chosen regression loss for continuous labels; the anchor means are reused as prediction anchors through a tied readout, so the declared geometry is shared by compression and prediction. Since the geometric prior remains a label-conditional reference, the CEB certificate of Eq. (1) carries over unchanged and every member of $\mathcal{G}$ inherits the guarantees of Section 2.2.

Instances: OPB and EPB. For classification, OPB evaluates a prior encoder on all class labels and uses its thin polar factor Q_θ . If U_θ is the resulting all-class matrix, then

$$Q_\theta = U_\theta(U_\theta^\top U_\theta)^{-1/2}, \quad Q_\theta^\top Q_\theta = I_K, \quad (6)$$

and we assume, monitoring rather than imposing it, that the monitored matrix has full column rank. Writing $q_{\theta,k}$ for the k -th column of the frame Q_θ , the anchors and their fixed geometry are

$$p_\theta^{\text{OPB}}(z|Y = k) = \mathcal{N}(aq_{\theta,k}, \tau^2 I), \quad (aq_i)^\top (aq_j) = a^2 \mathbf{1}\{i = j\}, \quad \|aq_i - aq_j\| = \sqrt{2}a \ (i \neq j). \quad (7)$$

The anchor-energy scores are the Gaussian Bayes scores

$$s_k(z) = -\frac{\|z - aq_{\theta,k}\|^2}{2\tau^2} + \log \pi_k, \quad (8)$$

With uniform priors, this induces an orthogonal classifier whose scale is fixed by the prior (Appendix B.2). Detailed polar construction, initialization, the covariance specification, and the relation to prototype/ETF methods are given in Appendices B.2 and D.

For regression, EPB uses a one-dimensional continuous label, min-max normalized once on the training set as $\tilde{y}$ with the statistics frozen. Let $W \in \mathbb{R}^{d \times 1}$ be a trainable direction with per-step normalization $u = W/\|W\|$; the isometric prior is

$$p_\theta^{\text{EPB}}(z|y) = \mathcal{N}(\rho \tilde{y} u, \tau^2 I), \quad \|m_\theta(y) - m_\theta(y')\| = \rho |\tilde{y} - \tilde{y}'|, \quad (9)$$

so latent distances equal ρ times normalized label distances exactly. Order and magnitude of label differences are encoded with equality, which makes it a member of $\mathcal{G}$ (Eq. (4) with $\rho_{\min} = \rho_{\max} = \rho$). Why this exactness is automatic, its trade-off against equal anchor norms, and the multidimensional-label extension are given in Appendix B.2. The prediction head is the tied projection $\hat{y} = u^\top z / \rho$, the regression analogue of the anchor-energy classifier: no free direction or scale can bypass the prior.

2.2 ALIGNMENT, SEPARATION, AND GEOMETRY TRANSFER

This subsection traces how the objective of Eq. (5) shapes the learned posterior: Theorem 2.1 upper-bounds the posterior's anchor mismatch, Theorem 2.2 lower-bounds the separation of posterior centers and introduces the condition under which the bound is informative, and Corollary 2.3 combines the two into the end-to-end β -dependent transfer guarantee. The guarantees certify geometry transfer; they are not optimality statements and do not compare the constrained geometry with unconstrained references. Throughout, $\Delta(y, y')$ denotes the declared lower bound on the anchor

pair distance: $\Delta(y, y') = \Delta$ for distinct discrete labels (Eq. (3)) and $\Delta(y, y') = \rho_{\min} d_Y(y, y')$ for continuous labels (Eq. (4)). Define

$$\mathcal{R}(w) = \mathbb{E}[\ell(c_\psi(Z), Y)], \quad \mathcal{K}(w) = \mathbb{E} \text{KL}(q_\phi(z|X) \| p_\theta(z|Y)), \quad (10)$$

and

$$\mathcal{D}(w) = \frac{\mathbb{E} \|\mu_\phi(X) - m_\theta(Y)\|^2}{2\tau^2}. \quad (11)$$

Theorem 2.1 (Comparator-dependent bound on anchor mismatch). *Let $\beta > 0$, $\varepsilon \geq 0$, and $\ell \geq 0$. If $\hat{w}$ is ε -optimal for $\mathcal{R} + \beta\mathcal{K}$, then for every feasible comparator w_0 in the same family (same encoder, posterior, prior, and prediction-head parameterization),*

$$\mathcal{D}(\hat{w}) \leq \mathcal{K}(\hat{w}) \leq \mathcal{K}(w_0) + \frac{\mathcal{R}(w_0) + \varepsilon}{\beta}. \quad (12)$$

The bound certifies alignment to whatever comparator is prespecified: its KL $\mathcal{K}(w_0)$ is the comparator-specific certificate floor, a property of the chosen comparator rather than an intrinsic dataset quantity, and the risk term controls how fast the right-hand certificate bound approaches that floor as β grows. The floor’s size and reachability are examined empirically in Secs. 3.3 and 3.4. Equivalently, to certify a target mismatch level d_* , one needs a comparator with $\mathcal{K}(w_0) < d_*$ and $\beta \geq (\mathcal{R}(w_0) + \varepsilon)/(d_* - \mathcal{K}(w_0))$.

Theorem 2.2 (Mismatch-conditioned posterior-center separation). *Suppose $\mathbb{E} \|\mu_\phi(X)\|^2 < \infty$ and, for a label value y , let*

$$\bar{\mu}(y) = \mathbb{E}[\mu_\phi(X) | Y = y], \quad D(y) = \frac{\mathbb{E} \|\mu_\phi(X) - m_\theta(y)\|^2 | Y = y}{2\tau^2}. \quad (13)$$

Then, for every pair of distinct discrete labels and for $P_Y \otimes P_Y$ -almost every continuous pair (y, y') ,

$$\begin{aligned} \|\bar{\mu}(y) - \bar{\mu}(y')\| &\geq \|m_\theta(y) - m_\theta(y')\| - \sqrt{2\tau^2 D(y)} - \sqrt{2\tau^2 D(y')} \\ &\geq \Delta(y, y') - \sqrt{2\tau^2 D(y)} - \sqrt{2\tau^2 D(y')}. \end{aligned} \quad (14)$$

The first-level bound is positive exactly when

$$\Gamma_{yy'} := \frac{\sqrt{2\tau^2 D(y)} + \sqrt{2\tau^2 D(y')}}{\|m_\theta(y) - m_\theta(y')\|} < 1, \quad (15)$$

while the declared-margin level is positive exactly when the measured correction terms stay below the declared bound, $\sqrt{2\tau^2 D(y)} + \sqrt{2\tau^2 D(y')} < \Delta(y, y')$, which in the continuous case fails most readily for nearby label pairs. For OPB the two conditions coincide, since the anchor pair distance is the constant $\sqrt{2}a$ of Eq. (7): $\Delta(y, y') = \sqrt{2}a$ and, with equal mismatch D , positivity holds exactly for $D < a^2/(4\tau^2)$.

Corollary 2.3 (Beta-dependent transfer of anchor separation). *Fix the prior family, a feasible comparator w_0 , and an ε_β -optimal solution $\hat{w}_\beta$ for each $\beta > 0$, with $\sup_\beta \varepsilon_\beta < \infty$. Define*

$$B_\beta := \mathcal{K}(w_0) + \frac{\mathcal{R}(w_0) + \varepsilon_\beta}{\beta}. \quad (16)$$

For any label set S and mismatch level $\bar{D}_\beta \geq \sup_{y \in S} D(y)$,

$$\|\bar{\mu}(y) - \bar{\mu}(y')\| \geq \Delta(y, y') - 2\sqrt{2\tau^2 \bar{D}_\beta} \quad \text{for distinct } (y, y') \in S^2. \quad (17)$$

For discrete labels, suppose $\pi_k \geq \pi_{\min} > 0$ for all considered classes; then $\bar{D}_\beta = B_\beta/\pi_{\min}$ covers every class (Theorem 2.1 with the tower property), so Eq. (17) holds for every pair of distinct classes. For continuous labels, for any $\xi \in (0, 1)$, $\bar{D}_\beta = B_\beta/\xi$ covers a set $S_{\beta, \xi}$ of label mass at least $1 - \xi$ (Markov’s inequality), so Eq. (17) holds for $P_Y \otimes P_Y$ -almost every pair in $S_{\beta, \xi}^2$.

With a fixed comparator satisfying $\mathcal{K}(w_0) = 0$ and the remaining quantities bounded while β varies, the certified separation deficit is $O(\beta^{-1/2})$; with $\mathcal{K}(w_0) > 0$, the certificate approaches the comparator-dependent non-zero upper bound $\mathcal{K}(w_0)$. These rate statements concern the certificate only: they do not claim that the realized mismatch $\mathcal{D}(\hat{w}_\beta)$ or the realized center distances converge at this rate.

3 EXPERIMENTS

The evaluation addresses five research questions, each answered by the sections noted below.

(RQ1) End-to-end utility. Does GPB improve the operating point across dataset–backbone settings, and which design factor carries the gain? The main results (Sec. 3.2) report the former, and the paired factor-wise attribution ladder (Sec. 3.6) isolates the latter.

(RQ2) Comparator floor. Does the trained posterior attain the theoretically predicted ambiguity floor and never leave the explicit comparator’s reachable set? The controlled label-noise study varies a known floor from zero to 11.2 nats and checks the comparator gate (Sec. 3.3).

(RQ3) Geometry transfer and certification. How much of the declared prior geometry reaches the learned posterior, and does the separation guarantee survive the measured mismatch? The posterior alignment and separation study measures the transfer, and the geometry-transfer study verifies the posterior-side structure (Secs. 3.4 and 3.9).

(RQ4) Compression behavior. What do the methods compress and discard along the β sweep, and how does the anchor scale trade accuracy against compression? The compression curves and the information-plane estimates read both (Secs. 3.7 and 3.8).

(RQ5) Robustness. Does the compressed geometric operating point reduce targeted adversarial success without giving up clean accuracy? The white-box PGD study answers it (Sec. 3.5).

3.1 EXPERIMENTAL SETUP

We evaluate twelve dataset–backbone settings over nine public datasets: MNIST, ImageNet-100, Cora, IMDb, AG News, California Housing, STS-B, ZINC-12k, and AgeDB, covering MLP, CNN, GCN, and RNN backbones. All methods use the same data splits (60/20/20 train/validation/test for most datasets), backbone, Adam training, early stopping, validation-based selection, and ten seeds per configuration. The baselines are Base, VIB, SVIB, NIB, CEB, supervised β -DVCCA, and the external non-IB baselines NCBd (unit-hypersphere prototypes with a simplex-ETF contrastive optimum) and AdaCap (contrastive loss with a closed-form readout); GPB is OPB on classification rows and EPB on regression rows. The main comparison is single-dimensional in tuned hyperparameters: the plain backbone has no method-specific hyperparameters, every β -weighted method is swept over the same 30-point β grid (six decades, five values per decade), and NCBd and AdaCap use 30-value grids of their own. GPB’s anchor scale a (resp. isometric scale ρ) is fixed and selected on the five-value grid $\{1, 3, 6, 9, 12\}$, so GPB searches a two-dimensional grid of $6 \times 5 = 30$ configurations, matching the baselines’ budget. For each method and setting, the configuration with the best mean validation metric is selected (accuracy on classification settings, R^2 on regression settings). The reported results are means with standard deviations over seeds, and paired differences use 95% bootstrap confidence intervals. The complete experimental setup is in Appendix B.1.

3.2 MAIN RESULTS

Table 1 reports the best validation-selected configuration of every baseline and GPB variant; the complete standard deviations and paired confidence intervals are in Appendix C.1; per-method configuration details are in Appendix B.1. The ranking summary is dominated by GPB: it has the lowest mean rank (1.58 versus 4.08 for the best plain baseline) and is best or tied-best in 8/12 settings. The gains are largest where the task has margin to resolve them: on ImageNet-100 (CNN) GPB leads CEB by 3.11 points and on Cora by 1.85, while the saturated settings are flat (MNIST CNN and AG News, where all methods sit within 0.6 points of the best) and the small-data regressions are essentially tied with the best baselines (Housing within 0.0006 R^2 of VIB; on STS-B the plain backbone remains best). Among the external non-IB baselines, AdaCap is competitive on regression (best on ZINC), while NCBd’s mean rank (5.43) places it behind the plain backbone on classification. Across modalities, the same construction improves the operating point where improvement is available and costs nothing elsewhere.

Table 1: Test accuracy (%) on classification tasks and test R^2 on regression tasks (mean over runs); the best entry per dataset is bolded and the second-best underlined. Dashes mark the settings where a method does not apply.

Dataset	Backbone	Base	VIB	SVIB	NIB	CEB	DVCCA	NCBD	AdaCap	GPB
MNIST	MLP	98.33	98.28	98.15	98.27	<u>98.43</u>	98.24	97.95	–	98.54
MNIST	CNN	99.13	99.28	99.23	99.22	<u>99.30</u>	99.29	99.18	–	99.32
ImageNet-100	MLP	83.12	82.84	83.02	82.35	<u>82.77</u>	82.99	<u>83.64</u>	–	84.40
ImageNet-100	CNN	<u>83.40</u>	82.94	83.11	82.91	82.97	83.21	<u>80.56</u>	–	86.08
Cora	GCN	<u>87.68</u>	86.83	86.46	86.35	86.27	86.90	87.34	–	88.12
IMDb	RNN	84.85	85.50	85.99	86.14	85.55	85.69	<u>86.69</u>	–	86.84
AG News	RNN	92.94	92.72	92.86	<u>92.91</u>	92.83	92.78	<u>92.37</u>	–	92.87
Cal. Housing	MLP	0.7618	0.7901	<u>0.7897</u>	0.7862	0.7894	0.7878	–	0.7758	0.7895
STS-B	RNN	0.2899	0.2719	0.2697	0.2417	0.2668	0.2713	–	0.1613	<u>0.2762</u>
ZINC	GCN	<u>0.6348</u>	0.6339	0.6325	0.6178	0.6257	0.6294	–	0.6770	0.6344
AgeDB	MLP	0.2180	0.3173	0.3385	0.2710	0.3317	0.3180	–	<u>0.3482</u>	0.3536
AgeDB	CNN	0.3914	0.3342	0.3323	0.3337	0.3882	0.3205	–	<u>0.4684</u>	0.4700
Mean rank		4.08	4.83	4.67	6.00	5.00	5.00	5.43	4.00	1.58
Best or tied-best		2	1	0	0	0	0	0	1	8

Table 2: Controlled label-noise floor (mean $\pm$ std over seeds); max excess is $\max_{\beta}(\hat{D} - \delta_{\text{noise}})$, and the last column is the fitted objective minus the explicit comparator objective at $\beta = 10$.

r_n	δ_{noise}	$\hat{D}(\beta=10^{-3})$	$\hat{D}(\beta=10)$	max excess	$\hat{L} - L_{\text{comp}}$
0	0.000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.000	0.000 $\pm$ 0.000
0.1	3.400	3.4142 $\pm$ 0.0002	3.4018 $\pm$ 0.0000	0.014	−0.005 $\pm$ 0.004
0.2	6.400	6.4225 $\pm$ 0.0005	6.4067 $\pm$ 0.0002	0.023	−0.053 $\pm$ 0.008
0.4	11.200	11.2324 $\pm$ 0.0016	11.2172 $\pm$ 0.0003	0.032	−0.191 $\pm$ 0.016

3.3 CONTROLLED LABEL NOISE

Theorem 2.1’s certificate assumes ε -optimality of the fitted solution, which non-convex training does not guarantee; the controlled experiment checks this premise and the certificate’s tightness where the comparator floor is analytically known. It isolates the comparator floor by making the label-ambiguity term of the mismatch bound analytically known, and verifies three claims: the noise-free case realizes the zero floor exactly, the measured mismatch tracks the analytic floor at every noise level, and the fitted objective never leaves the explicit comparator’s reachable set. The construction uses $K = 10$, $X = e_C$, and symmetric label noise $r_n \in \{0, 0.1, 0.2, 0.4\}$ with OPB $(a, \tau) = (6, 1)$. The analytic ambiguity floor (Appendix C.2) grows from 0 to 11.2 nats as r_n runs over the noise grid. The explicit comparator takes posterior mean $h_0(x) = a \sum_k \eta_k(x) q_k$ and covariance $\tau^2 I$, where $\eta_k(x) = P(Y = k \mid X = x)$; its KL equals $\delta_{\text{noise}}(r_n)$ exactly (Proposition A.2), and its anchor-energy risk is $C_{\text{comp}}(r_n)$. The training protocol (population-weighted objective, warm start, Monte Carlo risk estimation) is in Appendix C.2. Whenever the fitted objective satisfies $\hat{L} \leq L_{\text{comp}} = C_{\text{comp}} + \beta \delta_{\text{noise}}$, Theorem 2.1 certifies $\hat{D} \leq \delta_{\text{noise}} + C_{\text{comp}}/\beta$ directly. Thus the known-noise experiment tracks the predicted floor; the sensitivity controls and their comparator-gate checks are deferred to Appendix C.2.

Table 2 reads the transition from the zero to the non-zero floor column by column. The zero-noise row reproduces the exact-realizability limit of Corollary A.3 to the stored precision. For $r_n > 0$ the measured mismatch tracks the analytic floor at both ends of the β grid: at $\beta = 10$ it matches the floor to within 0.02 nats at every noise level, while at the smallest β it sits above the floor by an excess that grows with the noise level (from 0.014 to 0.032 nats) and declines monotonically with compression, without implying an exact $1/\beta$ law. The last column closes the loop: the fitted objective satisfies $\hat{L} \leq L_{\text{comp}}$ at every reported point, and the margin over the explicit comparator grows with the noise level (from −0.005 to −0.191), so the trained posterior moves farther inside the comparator’s reachable set exactly where the floor is largest.

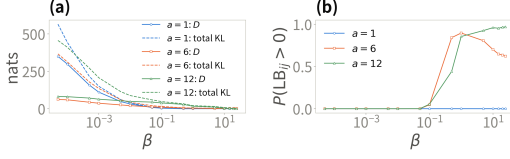


Figure 1: Alignment and separation diagnostics over the ImageNet-100 compression sweep. (a) Anchor mismatch $\hat{D}$ (solid) and total KL (dashed) against β , by anchor scale. (b) Fraction of class pairs with a positive separation lower bound.

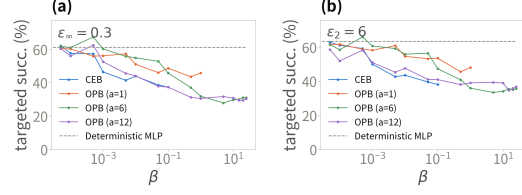


Figure 2: Targeted white-box PGD on MNIST (MLP): targeted success rate against β under (a) L_∞ , $\varepsilon = 0.3$ and (b) L_2 , $\varepsilon = 6$. Curves are shown only where stochastic clean accuracy exceeds 90%.

3.4 POSTERIOR ALIGNMENT AND SEPARATION

This section pursues two checks on the learned posterior: how much of the declared anchor geometry it inherits, and whether the separation guarantee survives under the measured mismatch. All statistics are first computed within each seed and then aggregated across seeds, and class or bin pairs are not treated as independent replicates. For test examples of class k , let $\hat{D}_k = \frac{1}{n_k} \sum_{n: y_n=k} \|\mu_\phi(x_n) - aq_{\theta,k}\|^2 / (2\tau^2)$ and $\hat{\mu}_k = \frac{1}{n_k} \sum_{n: y_n=k} \mu_\phi(x_n)$, the sample counterparts of $D(y)$ and $\bar{\mu}(y)$ in Theorem 2.2 ($\tau = 1$). Its two steps read, at sample level, as the per-class Jensen check $\|\hat{\mu}_k - aq_{\theta,k}\| \leq \sqrt{2\tau^2 \hat{D}_k}$ and the per-pair separation bound

$$\widehat{\text{LB}}_{ij} = \sqrt{2}a - \sqrt{2\tau^2 \hat{D}_i} - \sqrt{2\tau^2 \hat{D}_j}, \quad (18)$$

evaluated without clipping.

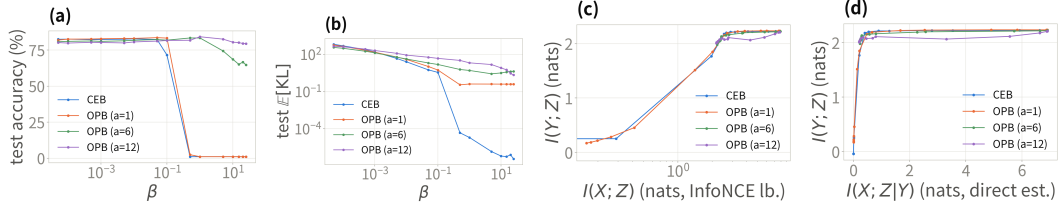
On ImageNet-100, the two panels read the condition and its certified consequence together (Fig. 1). Mismatch falls with compression: from 62.3 nats at $\beta = 5 \cdot 10^{-5}$ to 2.54 at $\beta = 25$ for $a = 6$, with the total KL co-plotted, and the ordering across scales reverses over the same range. At weak compression the smallest frame is farthest from its anchors (349.8, against 82.1 and 62.3), while at $\beta = 25$ the $a = 12$ frame is tightest among the separated frames (0.56 against 2.54). The positive-bound fraction rises exactly where the corrections shrink: $a = 12$ climbs to 96.9% at $\beta = 25$, $a = 6$ peaks at 89.5% near $\beta = 1$ and settles at 62.3%, and $a = 1$ never certifies, its corrections exceeding the declared margin at every β as its centers collapse under strong compression. The bound and slack readings, the decoupling scatter, and the regression panels are collected in Appendix C.3.

3.5 ADVERSARIAL ROBUSTNESS

We use the targeted MNIST PGD protocol of CEB with EOT-10 gradients and 10-sample stochastic predictions; each curve is shown only where the stochastic clean accuracy exceeds 90% (Fig. 2). Within that valid range, targeted success falls monotonically with compression: every curve starts at the deterministic baseline’s level ($\approx 61\%$ under L_∞ , $\approx 63\%$ under L_2) and declines by roughly a third to a half by the end of its valid range. The models differ in how far that range extends. CEB’s curve ends at $\beta = 0.1$, where its success (37.4%/38.1%) is matched by OPB $a = 12$ (37.4%/41.0%); beyond it CEB collapses toward chance accuracy, where near-zero success is trivial rather than a robustness gain, so no later point is plotted. OPB $a = 1$ ’s curve ends at $\beta = 1$, where its clean sampled-path accuracy drops below the 90% validity threshold. OPB $a = 6, 12$ stay valid over the whole grid and continue to fall, reaching 31.1%/30.5% under L_∞ and 35.7%/36.8% under L_2 at $\beta = 25$, roughly 30 points below the deterministic baseline under L_∞ and about 27 under L_2 , with clean accuracy still $\approx 98\%$; the curves plateau from $\beta \geq 5$ onward, so the gain sits in the compressed operating point rather than in β alone. Transfer attacks are reserved for the gradient-masking check in Appendix C.4.

Table 3: Factor-wise paired differences (accuracy points on MNIST and ImageNet-100; 95% bootstrap percentile CIs; positive values favor the first term).

factor	comparison	MNIST	ImageNet-100
readout	GPB – GPB-L	+0.02 [−0.05, 0.08]	+0.82 [0.34, 1.17]
readout (free geometry)	CEB-energy – CEB	−0.21 [−0.30, −0.12]	−1.19 [−1.50, −0.96]
geometry	OPB-FS – CEB-energy	+0.32 [0.26, 0.40]	+2.67 [2.45, 2.92]
fixed scale	GPB – OPB-FS	−0.09 [−0.16, −0.02]	−0.04 [−0.07, −0.01]
frame trainability	GPB – OPB-RV	+0.10 [0.05, 0.15]	+0.83 [0.29, 1.19]
IB increment	GPB – NCM-ortho	+0.34 [0.22, 0.45]	+1.89 [1.41, 2.36]
prototype structure	NCM-ortho – NCM-learn	−0.06 [−0.20, 0.11]	+0.16 [−0.49, 0.82]
prototype readout alone	NCM-learn – Base	−0.16 [−0.28, −0.05]	−0.96 [−1.30, −0.69]

Figure 3: Compression behavior across two views and two tasks. (a)–(b) ImageNet-100 classification sweep (MLP): test accuracy and realized conditional KL against β . (c)–(d) MNIST information-plane and certificate-plane estimates.

3.6 FACTOR-WISE ABLATIONS

The main comparison changes several factors simultaneously, so we also report a paired intervention ladder that switches five design factors independently, namely the prior geometry, the anchor scale, the readout, the IB term itself, and the trainability of the prior frame, plus two auxiliary contrasts on the prototype readout itself. Every variant is trained on the three mechanism settings (MNIST, ImageNet-100, and Cal. Housing, all with MLP backbones) at its best validation-selected configuration, and all comparisons are paired by seed. Each row of Table 3 pairs two variants that differ in the factor named in its first column; the variant definitions and the complete means are in Appendix C.5.

The paired contrasts (Table 3; accuracy-point differences) rank the factors on ImageNet-100: geometry is the largest (+2.67, OPB-FS over CEB-energy), followed by the IB term (+1.89, GPB over NCM-ortho, which additionally contains frame trainability) and the tied readout (+0.82, GPB over GPB-L), while fixing the anchor scale costs essentially nothing (−0.04, GPB over OPB-FS). Two patterns matter as much as the ranking. First, the readout contrast flips sign with the accompanying geometry: on the free-geometry side the energy readout costs 1.19 points (CEB-energy below CEB), so restricting the readout helps only when the geometry it ties to is the constructed one. Second, the prototype machinery contributes nothing on its own: NCM-learn trails the plain backbone by 0.96 points, and fixing its prototypes to the orthogonal frame without the IB term is a null (+0.16, CI spanning zero), so neither the energy form nor the frozen geometry is the source of the gain. On MNIST all contrasts shrink to within 0.35 points, so the ladder resolves factors only where the task margin is large enough. The frozen-prior ablation (Appendix C.5) shows the trainable frame carries task benefit on classification: GPB exceeds the variant with all prior parameters frozen by +0.10 on MNIST and +0.83 on ImageNet-100 (Table 3, frame trainability). These interventions change the fitted training problem and therefore support factor-wise attribution, not a unique causal decomposition.

3.7 COMPRESSION SWEEPS

This subsection shows the classification sweep on ImageNet-100 (MLP): test accuracy and realized conditional KL against β , for CEB and OPB at anchor scales $a \in \{1, 6, 12\}$ (Fig. 3a,b); the California Housing regression companion is in Appendix C.6.

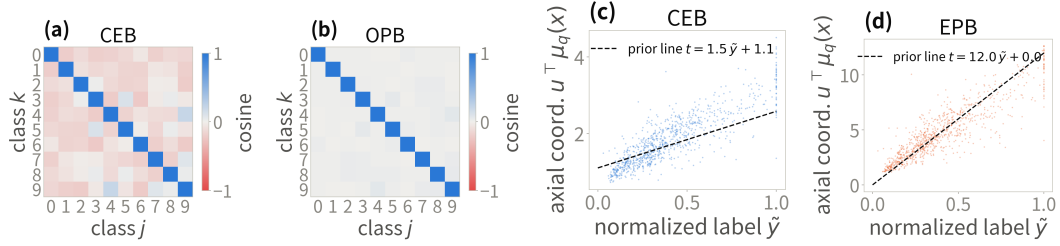


Figure 4: Posterior geometry: class-center Gram matrices for (a) CEB and (b) OPB, and axial coordinates for (c) CEB and (d) EPB.

Larger anchor scales retain the high-accuracy plateau farther into strong compression: $a = 12$ keeps 79.4% accuracy at $\beta = 25$, where CEB has already left its valid high-accuracy range and $a = 1$ collapses, while at weak compression larger scales pay a one-to-three point cost. The trade-off reverses under stronger compression, where the $a = 12$ scale is the most accurate configuration on the grid ($\beta \geq 1$). The KL panel shows where the two objectives spend the penalty: CEB’s KL falls from 3.6 at $\beta = 0.1$ to $\sim 10^{-5}$ at $\beta = 0.5$ because its learned reference shrinks to absorb the term, whereas OPB’s KL stays anchored at the geometric scale of its fixed prior (48 at $\beta = 0.1$, 21 at $\beta = 1$) and is dominated by anchor alignment, so the structure is retained rather than absorbed. Together the curves make the anchor-scale/compression trade-off visible across the useful operating range. The complementary scale-axis view, reading the full five-point scale grid $\{1, 3, 6, 9, 12\}$ at three fixed β levels, is reported in Appendix C.7.

3.8 INFORMATION-PLANE DIAGNOSTIC

The MNIST information-plane estimates provide a second view of the compression behavior (Fig. 3c,d). They are estimator-level diagnostics; the matched table and estimator details are in Appendix C.8, with the Housing companion in Appendix C.9.

The two planes show what each method discards as compression deepens. In the information plane (Fig. 3c, $\log I(X; Z)$ axis), CEB follows the classic corner: compression and task information fall together, from $I(X; Z) = 2.54$ nat at $\beta = 0.1$ to 1.96 nat at $\beta = 0.5$, and by $\beta = 1$ both are near zero. OPB with $a = 6, 12$ compresses from ≈ 7.9 to ≈ 2.2 nat but never below it, with $I(Y; Z)$ flat just below the $H(Y) = \ln 10$ ceiling, so compression stops at the task-information level, consistent with the aligned-model Fano lower bound of Corollary A.4. OPB with $a = 1$ tracks $a = 6, 12$ until $\beta = 0.5$ and then collapses toward CEB’s corner, with $I(Y; Z)$ falling to 0.17–0.45 nat. The certificate plane (Fig. 3d) reads the same contrast as where each trajectory ends: CEB and OPB with $a = 1$ reach zero residual together with the task-information collapse, whereas $a = 6, 12$ terminate at full task height on a stable nonzero residual plateau of 0.20–0.46 nat.

3.9 GEOMETRY TRANSFER

At matched compression $(\beta, a, \rho) = (0.1, 12, 12)$, the posterior geometry is read directly off the class-center cosine Gram matrices (Fig. 4a,b; Table 4): OPB’s Gram is visually an identity matrix, whereas CEB’s shows visible off-diagonal structure with no clean pattern. The posterior thus inherits the constructed orthogonal frame, which is exactly what the zero-parameter tied readout relies on. On Housing (Fig. 4c,d), the axial scatter shows the same contrast: EPB’s points hug its prior line, whereas CEB’s cloud deviates from its prior line’s direction and its learned axis scale swings across runs: its geometry is neither aligned with its reference nor stable across runs. The tied-metric column of Table 4 shows the payoff: the tied heads attain 98.44% on MNIST against CEB’s 98.16%, and $R^2 = 0.778$ on Housing against 0.491.

Conclusion and limitations. The experiments form a consistent progression from construction to behavior: the baseline table establishes end-to-end utility across modalities; the factor interventions attribute the gain to the geometric prior; the controlled noise study matches the analytic ambiguity floor; and the mismatch, robustness, compression, information-plane, and geometry diagnostics show how the declared prior is reflected in the learned posterior at different operating points. Con-

Table 4: Posterior-geometry statistics: geometry concentration (mean absolute off-diagonal cosine of the class-center Gram; off-axis fraction for Housing), scale following (posterior center norm against the declared anchor scale; slope against ρ for Housing), tied metric (accuracy or R^2 under the tied head), and cross-seed stability (standard deviation of the concentration across runs).

Model	Geometry concentration	Scale following	Tied metric	Cross-seed stability
CEB (MNIST)	0.1205 ± 0.0026	5.14	98.16%	0.0201
OPB (MNIST)	0.0203 ± 0.0021	11.00	98.44%	0.0082
CEB (Housing)	0.0233	2.65 ± 0.59	$R^2 = 0.491$	0.593
EPB (Housing)	0.0002	9.22 ± 0.15	$R^2 = 0.778$	0.155

straining the conditional prior changes the geometry available to the KL, and the tied readout makes that geometry visible to prediction; the measured mismatch determines how much of the declared geometry transfers.

The results are qualified accordingly. The mismatch bound is comparator-relative: on a real benchmark its floor is determined by a prespecified comparator and its label, mean-approximation, and covariance terms rather than an intrinsic dataset constant, and the separation lower bound is informative only when the realized mismatch stays below the declared margin. The orthogonal frame fixes all pairwise anchor distances, so the framework does not claim to recover arbitrary semantic class similarities; the trainable frame is a rotation channel whose certificate-zero set contains full-retention rotated codes, so the framework pins the geometry of the code rather than the retained label information. Data-driven comparator certificates and tighter information estimates are natural next steps. Proofs and all supplementary evidence are collected in Appendices A–D.

AI USE STATEMENT

(This section is **required** and does not count toward the page limit.)

In this work, we used generative AI tools for drafting and editing portions of the code and of this manuscript. All AI-assisted content was manually reviewed and verified by the authors; experimental results and theoretical claims were produced and checked by our own code and analysis. We take responsibility for the final content of this work, including text, claims or artifacts produced with the aid of generative AI.

REPRODUCIBILITY STATEMENT

The code, hyperparameter grids, and evaluation protocol are described in Section 3.1 and Appendix B.1; the geometric prior maintains no EMA prototypes or batch statistics and the orthogonalization is exact wherever defined. All datasets are public, with preprocessing steps given in the release. Proofs of all statements are collected in the appendix. Statistical claims use the paired bootstrap confidence intervals and non-inferiority margin described in Appendix C.1.

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NOTATION

Each symbol has a single meaning throughout the paper; symbols confined to one statement or section are marked with their scope.

Symbol	Object	Meaning
X, Y	variables	input and label random variables
$Z (z, Z_{\text{align}})$	code	representation random variable, its realization, and the aligned code of the comparator
P_{XY}, P_Y	distributions	data distribution and label marginal
$\mathbb{E}, \text{KL}, \text{CE}, \text{Var}$	operators	expectation, KL divergence, cross-entropy, variance
$H(\cdot), I(\cdot; \cdot)$	information	entropy and mutual information
$\mathcal{N}(\mu, \Sigma)$	Gaussian	multivariate Gaussian distribution
$q_\phi(z x)$	posterior	reparameterized Gaussian encoder distribution
$\mu_\phi(x), \sigma_\phi^2(x)$	posterior mean / variance	learned diagonal posterior parameters
$p_\theta(z y)$	geometric prior	label-conditional Gaussian reference
$m_\theta(y)$	anchor mean	prior mean function (orthogonal anchors $a_{q_{\theta,k}}$ for OPB, isometric axis $\rho \tilde{y} u$ for EPB)
τ^2	prior variance	fixed isotropic prior covariance scale, $\tau^2 = 1$
a, ρ	scales	anchor scale (OPB) / isometric scale (EPB)
$Q_\theta, q_{\theta,k}$	frame	all-class polar frame and its k -th column (OPB); the same frame primitive of the m -dimensional EPB extension (Appendix B.2)
u	unit direction	EPB normalized trainable direction
W	direction matrix	EPB trainable direction before normalization
$u_0, q_{0,k}$	comparator frame	fixed unit direction / fixed orthogonal frame of the explicit comparator
$s_k(z)$	score	anchor-energy score of class k at representation z
$\tilde{y} (\tilde{Y})$	normalized label	min-max normalized target (and its random variable)
$\hat{y}$	prediction	tied projection of the regression head
$\pi_k, \pi_{\min}$	class priors	label probabilities and their minimum
K, d	dimensions	number of classes / latent dimension
d_Y	label metric	metric on the label space
β	KL weight	compression weight in the objective
ℓ, c_ψ	loss, readout	prediction loss and classifier/regression head
$w, \hat{w}, w_0$	models	trainable parameters, trained solution, comparator
$\mathcal{L}_{\text{GPB}}$	objective	GPB training objective of Eq. (5)
$\mathcal{R}(w), \mathcal{K}(w)$	risk, KL	prediction risk and expected KL of model w
$\mathcal{D}(w), D(y)$	mismatch	anchor mismatch at model / class level
B_β	budget	certificate budget of Corollary 2.3
$\bar{D}_\beta$	mismatch level	per-label level with $\bar{D}_\beta \geq \sup_{y \in S} D(y)$ on the considered label set (Corollary 2.3)
$\Delta, \bar{\Delta}, \Delta(y, y')$	declared separation	minimal/maximal anchor distance and the pair lower bound
$\rho_{\min}, \rho_{\max}$	Lipschitz constants	bi-Lipschitz bounds of the anchor map
$\Gamma_{yy'}$	tightness ratio	separation-tightness ratio of Theorem 2
$\mathcal{G}$	family	geometric prior family of the GPB framework
$\bar{\mu}(y)$	posterior center	$\mathbb{E}[\mu_\phi(X) Y = y]$
$\eta_k(x) (\eta(x))$	label probability	$\mathbb{P}(Y = k X = x)$ and its vector
$h_0(x)$	comparator mean	explicit comparator's posterior mean
$\nu(X)$	conditional mean	$\mathbb{E}[\tilde{Y} X]$ (EPB)
V_0	covariance mismatch	variance part of the comparator KL
A_Y	random anchor	anchor of the sampled label
C_G	Bayes risk	Bayes cross-entropy of the aligned comparator (proof of Corollary A.3)
ξ	coverage fraction	label mass coverage in Corollary 2.3
γ	CEB weight	Lagrangian weight of the CEB objective (Appendix A.3 discussion only)
$\delta_{\text{noise}}(r_n)$	ambiguity floor	analytic floor at noise rate r_n
r_n	noise rate	symmetric label noise rate (controlled experiment)
$\hat{D} (\hat{D}_k), \hat{\mu}_k$	empirical statistics	sample mismatch (per class) and per-class center
LB	empirical bound	sample separation lower bound

Symbol	Object	Meaning
$\widehat{L}, L_{\text{comp}}, C_{\text{comp}}$	objectives	fitted objective, explicit comparator's objective and risk (controlled experiment)
$P_{\text{pair}}, P_e, \bar{P}_e$	error probabilities	pairwise / overall error of the nearest-anchor classifier in the aligned model, and its displayed bound
Φ, h_2	functions	standard normal CDF and binary entropy (Fano corollary)
T	temperature	NCBD temperature (Appendix B.1)
λ	Tikhonov	AdaCap Tikhonov regularization (Appendix B.1)
δ_{NI}	margin	non-inferiority margin of the paired CIs (Appendix C.1)
B	resamples	bootstrap resample count (Appendices C.1, C.5)
$B_b, \bar{y}_b, h_b, \bar{\mu}_i$	bins	label bin, its representative label, its diameter, and its center (Appendix C.3)
$N_{\text{neg}}, \mathcal{L}_{\text{NCE}}$	InfoNCE	negative count and InfoNCE loss (Appendix C.8)
M	draws	Monte Carlo draw count per test point (Appendix C.8)
h	bandwidth	Gaussian-kernel bandwidth of the regression mixture (Appendix C.9)
$\hat{q}(z y)$	mixture	estimated class-conditional mixture density (Appendix C.9)
$\hat{y}_{\text{MC}}$	prediction	Monte Carlo prediction of the sampled posterior (Appendix C.9)
$\varepsilon (\varepsilon_\beta)$	tolerance	(β -dependent) optimality tolerance of the training objective
$\varepsilon_\infty, \varepsilon_2$	attack budgets	adversarial perturbation bounds (Appendix C.4)

A PROOFS AND THEORETICAL ANALYSIS

A.1 PROOFS OF THE MAIN GUARANTEES

Lemma A.1 (OPB KL decomposition). *For a diagonal posterior covariance and the prior of Eq. (7), the standard diagonal-Gaussian KL formula reads*

$$\text{KL}(q_\phi(z|x) \| p_\theta^{\text{OPB}}(z|y)) = \frac{1}{2\tau^2} \|\mu_\phi(x) - a_{q_\theta, y}\|^2 + \frac{1}{2} \sum_{r=1}^d \left[\frac{\sigma_{\phi, r}^2(x)}{\tau^2} - 1 - \log \frac{\sigma_{\phi, r}^2(x)}{\tau^2} \right], \quad (19)$$

a class-anchor mismatch term plus a variance mismatch term that is non-negative and vanishes exactly at $\sigma_{\phi, r}^2(x) = \tau^2$ for all r .

Proof of Theorem 2.1. The Gaussian KL of Lemma A.1 is the sum of its mean and variance terms, the latter non-negative, so with the fixed prior variance $\tau^2 = 1$,

$$\text{KL}(q_\phi(z|x) \| p_\theta(z|y)) \geq \frac{1}{2\tau^2} \|\mu_\phi(x) - m_\theta(y)\|^2 = \frac{\|\mu_\phi(x) - m_\theta(y)\|^2}{2},$$

and hence $\mathcal{D}(\hat{w}) \leq \mathcal{K}(\hat{w})$. For any feasible w_0 in the same family, ε -optimality of $\hat{w}$, $\ell \geq 0$, and $\beta > 0$ give

$$\beta \mathcal{K}(\hat{w}) \leq \mathcal{R}(\hat{w}) + \beta \mathcal{K}(\hat{w}) \leq \mathcal{R}(w_0) + \beta \mathcal{K}(w_0) + \varepsilon.$$

Division by β proves Eq. (12). $\square$

Proof of Theorem 2.2. We first bound each posterior center's distance to its own anchor, then combine the two endpoint bounds with the triangle inequality. For the first step, convexity of the norm and Jensen's inequality give

$$\begin{aligned} \|\bar{\mu}(y) - m_\theta(y)\| &= \|\mathbb{E}[\mu_\phi(X) - m_\theta(y) \mid Y = y]\| \\ &\leq \mathbb{E}[\|\mu_\phi(X) - m_\theta(y)\| \mid Y = y] \\ &\leq \left(\mathbb{E}[\|\mu_\phi(X) - m_\theta(y)\|^2 \mid Y = y] \right)^{1/2} = \sqrt{2\tau^2 D(y)}, \end{aligned}$$

where the first inequality is Jensen's inequality on the convex norm, the second Jensen's inequality on the concave square root (equivalently Cauchy-Schwarz), the last equality is the definition of

$D(y)$ in Eq. (13), and $\mathbb{E}\|\mu_\phi(X)\|^2 < \infty$ makes $\bar{\mu}(y)$ well defined. For the second step, decompose the center difference as a three-leg path through the anchors,

$$\bar{\mu}(y) - \bar{\mu}(y') = (\bar{\mu}(y) - m_\theta(y)) + (m_\theta(y) - m_\theta(y')) + (m_\theta(y') - \bar{\mu}(y')),$$

and apply the triangle inequality in its reverse form: the middle leg (the anchor distance) is at least the total length minus the two endpoint legs, so together with the bound above,

$$\begin{aligned} \|\bar{\mu}(y) - \bar{\mu}(y')\| &\geq \|m_\theta(y) - m_\theta(y')\| - \sqrt{2\tau^2 D(y)} - \sqrt{2\tau^2 D(y')} \\ &\geq \Delta(y, y') - \sqrt{2\tau^2 D(y)} - \sqrt{2\tau^2 D(y')}, \end{aligned}$$

the last step replacing the anchor distance by its declared lower bound $\Delta(y, y')$. These are the two levels of Eq. (14). The first level is positive exactly when $\Gamma_{yy'} < 1$ and the declared-margin level exactly when $\sqrt{2\tau^2 D(y)} + \sqrt{2\tau^2 D(y')} < \Delta(y, y')$; for OPB the anchor pair distance is the constant $\sqrt{2}a$ of Eq. (7), so the declared bound satisfies $\Delta(y, y') = \sqrt{2}a$ and the two conditions coincide. With equal mismatch D the common condition

$$\Gamma_{yy'} = \frac{2\sqrt{2\tau^2 D}}{\sqrt{2}a} < 1 \quad \Longleftrightarrow \quad D < \frac{a^2}{4\tau^2}.$$

For continuous labels the Jensen inequality holds pointwise for P_Y -a.e. y (regular conditional expectation), and intersecting the two measure-one sets of valid y 's yields the $P_Y \otimes P_Y$ -almost-everywhere pair statement. $\square$

Remark A.1 (Finite-sample estimation of the continuous case). On finite samples the regular conditional expectations of the continuous case are estimated by binning the label axis into intervals B_i with representative labels $\bar{y}_i$ and centers $\bar{\mu}_i = \mathbb{E}[\mu_\phi(X) \mid Y \in B_i]$; the bin diameter adds a quantization error bounded by

$$\rho_{\max} \text{diam}(B_i) + \rho_{\max} \text{diam}(B_j)$$

(by the upper bound of Eq. (4)), which vanishes as the bins shrink. The separation statement itself is pointwise and needs no binning.

Proof of Corollary 2.3. Theorem 2.1 gives

$$\mathcal{D}(\hat{w}_\beta) \leq B_\beta.$$

For discrete labels, the tower property gives $\mathcal{D}(\hat{w}_\beta) = \sum_k \pi_k D(k)$, hence

$$D(k) \leq B_\beta / \pi_k \leq B_\beta / \pi_{\min};$$

for continuous labels, Markov's inequality on the nonnegative random variable $D(Y)$ gives

$$P_Y(D(Y) > B_\beta / \xi) \leq \xi.$$

In both cases every $y \in S$ satisfies $D(y) \leq \bar{D}_\beta$, and the declared-margin level of Theorem 2.2 then gives

$$\begin{aligned} \|\bar{\mu}(y) - \bar{\mu}(y')\| &\geq \Delta(y, y') - \sqrt{2\tau^2 D(y)} - \sqrt{2\tau^2 D(y')} \\ &\geq \Delta(y, y') - 2\sqrt{2\tau^2 \bar{D}_\beta}, \end{aligned}$$

the last step by monotonicity of the square root; this is Eq. (17). The rate statement follows directly from the definition of B_β : with $\mathcal{K}(w_0) = 0$ and $\sup_\beta \varepsilon_\beta < \infty$, $B_\beta = O(\beta^{-1})$ so the certified deficit is $O(\beta^{-1/2})$; with $\mathcal{K}(w_0) > 0$, $B_\beta \rightarrow \mathcal{K}(w_0)$ as $\beta \rightarrow \infty$. $\square$

Scope of the comparison argument. Theorem 2.1 applies to arbitrary P_{XY} and every feasible comparator in the fitted model class; it does not require $q(z|x) = p_\theta(z|y)$. Its usefulness depends on the comparator KL. Label ambiguity, limited encoder capacity, and covariance misspecification produce the non-zero floor made explicit in Proposition A.2. The stronger statement that this floor is zero still requires the assumptions of Corollary A.3: deterministic labels, exact posterior–prior matching, and a compatible head. We do not assume those exact-realizability premises for the real-data benchmarks. Their performance comparison evaluates utility, while their geometric diagnostics report realized mismatch and geometry. The controlled known-noise experiment instead evaluates the ambiguity term where its population value is analytic, including the deterministic distribution as its zero-floor special case.

A.2 DEFERRED STATEMENTS AND PROOFS

Proposition A.2 (Ambiguity–approximation decomposition). *For OPB classification with $\eta_k(x) = P(Y = k | X = x)$ and a comparator mean $h_0(x)$,*

$$\mathcal{K}(w_0) = \frac{a^2}{2\tau^2} \mathbb{E}[1 - \|\eta(X)\|_2^2] + \frac{1}{2\tau^2} \mathbb{E} \left\| h_0(X) - a \sum_k \eta_k(X) q_{0,k} \right\|^2 + V_0, \quad (20)$$

where V_0 is the covariance mismatch. For EPB regression,

$$\mathcal{K}(w_0) = \frac{\rho^2}{2\tau^2} \mathbb{E}[\text{Var}(\tilde{Y} | X)] + \frac{1}{2\tau^2} \mathbb{E} \| h_0(X) - \rho \nu(X) u_0 \|^2 + V_0. \quad (21)$$

Substituting either decomposition into B_β in Corollary 2.3 separates the comparator-dependent geometry transfer floor into label ambiguity, encoder mean approximation, and covariance mismatch.

Proof. For a comparator with posterior mean h_0 and diagonal covariance, Lemma A.1 splits the KL into its mean and variance terms,

$$\mathcal{K}(w_0) = \frac{1}{2\tau^2} \mathbb{E} \| h_0(X) - A_Y \|^2 + V_0,$$

where $A_Y := a q_{0,Y}$ is the random anchor of the sampled label. It remains to decompose the mean term. Conditional on X , orthonormality gives

$$\mathbb{E}[A_Y | X] = a \sum_k \eta_k(X) q_{0,k},$$

and, since every anchor is a unit vector and the frame is orthonormal,

$$\mathbb{E} \left[\| A_Y - \mathbb{E}[A_Y | X] \|^2 | X \right] = a^2 \left(\mathbb{E} [\| q_{0,Y} \|^2 | X] - \left\| \sum_k \eta_k(X) q_{0,k} \right\|^2 \right) = a^2 \left(1 - \sum_k \eta_k^2(X) \right).$$

The conditional bias–variance identity

$$\mathbb{E} \| h_0(X) - A_Y \|^2 = \mathbb{E} [\| A_Y - \mathbb{E}[A_Y | X] \|^2] + \mathbb{E} [\| h_0(X) - \mathbb{E}[A_Y | X] \|^2]$$

(the cross term vanishes by the definition of conditional expectation) then gives

$$\mathbb{E} \| h_0(X) - A_Y \|^2 = a^2 \mathbb{E} [1 - \|\eta(X)\|_2^2] + \mathbb{E} \left\| h_0(X) - a \sum_k \eta_k(X) q_{0,k} \right\|^2,$$

which proves Eq. (20). For EPB set $A_Y := \rho \tilde{Y} u_0$; then

$$\mathbb{E}[A_Y | X] = \rho \nu(X) u_0 \quad \text{and} \quad \mathbb{E} [\| A_Y - \mathbb{E}[A_Y | X] \|^2 | X] = \rho^2 \text{Var}(\tilde{Y} | X).$$

The same identity plus V_0 proves Eq. (21). $\square$

Corollary A.3 (Zero-floor exact-realizability special cases). *Under deterministic labels, exact posterior–prior matching, and a compatible Bayes head, $\mathcal{K}(w_0) = 0$ and Theorem 2.1 gives*

$$\mathcal{D}(\hat{w}) \leq (\ln K + \varepsilon)/\beta \quad (22)$$

for classification. For EPB regression,

$$\mathcal{D}(\hat{w}) \leq (\tau^2/\rho^2 + \varepsilon)/\beta. \quad (23)$$

Proof. For classification, the assumed comparator $q(z|x) = p_\theta(z|y)$ has $\mathcal{K}(w_0) = 0$. Its Bayes cross-entropy is

$$C_G = H(Y | Z_{\text{align}}) \leq \ln K,$$

since with shared covariance the determinant factors cancel and its posterior is the anchor-energy rule. Applying Eq. (12) gives Eq. (22). For regression the aligned comparator again has zero KL. Under the tied head its error is $(\tau/\rho)\eta$ with $\eta \sim \mathcal{N}(0, 1)$, whose standardized squared risk is τ^2/ρ^2 ; the same proposition gives Eq. (23). $\square$

Corollary A.4 (Label-information lower bound in the aligned Gaussian model). *Assume uniform classes and exact alignment $q(z|Y = k) = \mathcal{N}(a q_{\theta,k}, \tau^2 I_d)$, and let P_e be the error probability of the nearest-anchor classifier. Its pairwise error probability is*

$$P_{\text{pair}} = \Phi\left(-\frac{a}{\sqrt{2}\tau}\right), \quad P_e \leq \min\{1, (K-1)P_{\text{pair}}\}, \quad (24)$$

and whenever the displayed bound satisfies $\bar{P}_e \leq 1 - 1/K$, Fano’s inequality gives

$$I(Y; Z) \geq \log K - h_2(\bar{P}_e) - \bar{P}_e \log(K-1). \quad (25)$$

Conditionally on the aligned Gaussian model, the declared scale-to-noise ratio a/τ thus fixes how much label information the geometry alone can guarantee, the theoretical reference for the task-information plateaus of Section 3.8.

Proof. The orthonormal frame of Eq. (7) gives any two anchors distance $\sqrt{2}a$. For two Gaussians with the shared covariance $\tau^2 I_d$, the Bayes boundary is the perpendicular bisector of their centers, and the binary error is the Gaussian tail beyond the half-distance,

$$P_{\text{pair}} = \mathbb{P}(\mathcal{N}(0, 1) > a/(\sqrt{2}\tau)) = \Phi(-a/(\sqrt{2}\tau)).$$

The union bound over the $K-1$ competing classes gives the displayed bound $\bar{P}_e$. Fano’s inequality reads $H(Y|Z) \leq h_2(P_e) + P_e \log(K-1)$; the right-hand side is increasing in its argument on $[0, 1 - 1/K]$, so substituting $P_e \leq \bar{P}_e$ preserves the inequality, and $I(Y; Z) = H(Y) - H(Y|Z) = \log K - H(Y|Z)$ concludes. $\square$

B SETUP AND CONSTRUCTION DETAILS

B.1 COMPLETE EXPERIMENTAL SETUP

Scope and shared settings. We evaluate twelve dataset–backbone settings over nine public datasets: MNIST, ImageNet-100, Cora, IMDB, AG News, California Housing, STS-B, ZINC-12k, and AgeDB, covering MLP, CNN, GCN, and RNN backbones. All methods use the same data splits (60/20/20 train/validation/test with a fixed seed for most datasets, official splits kept where provided, e.g., Cora and ZINC-12k), backbone, Gaussian posterior with learned diagonal variance, bottleneck $d = 256$, Adam training, early stopping, and validation-based selection; GPB’s prior covariance is the fixed shared $\tau^2 I$ ($\tau^2 = 1$).

Training defaults and preprocessing. The shared training defaults are hidden sizes 512–256, dropout 0.2, learning rate 10^{-3} (Cora uses 10^{-2}), batch size 512, and a 200-epoch budget with early stopping (patience 20); dataset-specific defaults are likewise shared by every method on that dataset: STS-B uses a single-layer LSTM with hidden size 256 and frozen GloVe embeddings; AG News uses 50-dimensional PCA-reduced BERT features; ImageNet-100 uses 4×4 spatial pool-4 ResNet-50 features; the remaining dataset-specific preprocessing is given in the release. Every configuration is trained on the same ten seeds (0–9). Input features are standardized per dataset, and regression targets are min–max normalized to $[0, 1]$; both are fitted on the training split only, and reported MAE is rescaled to the original units.

Baselines and search-budget fairness. The baselines are Base, VIB, SVIB, NIB, CEB, supervised β -DVCCA, and the external non-IB baselines NCBD and AdaCap: NCBD trains unit-hypersphere features and prototypes with the NONL contrastive objective (simplex-ETF optimum), and AdaCap pairs a permutation contrastive loss with a Tikhonov closed-form readout, so the comparison spans geometric/prototype and contrastive non-IB mechanisms in addition to the IB variants. GPB is OPB on classification rows and EPB on regression rows. The main comparison equalizes the search budget: the plain backbone has no method-specific hyperparameters (one configuration); every β -weighted method, VIB, SVIB, NIB, CEB, and DVCCA, is swept over the same 30-point β grid (six decades, five values per decade); GPB searches the two-dimensional grid $\beta \in \{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1, 10\} \times a = \rho \in \{1, 3, 6, 9, 12\}$, also $6 \times 5 = 30$ configurations; NCBD and AdaCap use 30-value grids of their own. The same validation-based selection rule applies to every ablation variant (Appendix C.5). The per-method grids are summarized in Table 6.

Table 6: Per-method hyperparameter grids of the main comparison (Table 1); the best validation configuration is selected per method and setting.

Method	Method-specific hyperparameters	Configs.
Base	—	1
VIB, SVIB, NIB, CEB, DVCCA	$\beta \in \{1, 2, 4, 6, 8\} \times 10^k, k = -4, \dots, 1$	30
NCBD	temperature T : 30 log-spaced values in $[0.05, 1]$	30
AdaCap	Tikhonov $\lambda \in \{1, 2, 4, 6, 8\} \times 10^k, k = -2, \dots, 3$	30
GPB (OPB / EPB)	β grid above $\times a = \rho \in \{1, 3, 6, 9, 12\}$	$6 \times 5 = 30$

Table 7: Test accuracy (%) and test R^2 with standard deviations over runs: the plain backbone, the variational baselines, and NCBD (classification settings only); same configurations as Table 1.

Dataset	Backbone	Base	VIB	SVIB	NIB	NCBD
MNIST	MLP	98.33±0.12	98.28±0.11	98.15±0.13	98.27±0.07	97.95±0.23
MNIST	CNN	99.13±0.05	99.28±0.09	99.23±0.03	99.22±0.06	99.18±0.07
ImageNet-100	MLP	83.12±0.14	82.84±0.44	83.02±0.37	82.35±0.27	83.64±0.07
ImageNet-100	CNN	<u>83.40±0.13</u>	82.94±0.37	83.11±0.86	82.91±0.35	80.56±1.86
Cora	GCN	87.68±0.46	86.83±0.28	86.46±1.87	86.35±0.68	87.34±1.60
IMDb	RNN	84.85±0.68	85.50±0.91	85.99±0.30	86.14±0.92	86.69±0.52
AG News	RNN	92.94±0.15	92.72±0.17	92.86±0.21	92.91±0.22	92.37±0.37
Cal. Housing	MLP	0.7618±0.0238	0.7901±0.0049	0.7897±0.0058	0.7862±0.0035	—
STS-B	RNN	0.2899±0.0113	0.2719±0.0164	0.2697±0.0202	0.2417±0.0221	—
ZINC	GCN	0.6348±0.0044	0.6339±0.0033	0.6325±0.0026	0.6178±0.0099	—
AgeDB	MLP	0.2180±0.0409	0.3173±0.0101	0.3385±0.0286	0.2710±0.0235	—
AgeDB	CNN	0.3914±0.0194	0.3342±0.1894	0.3323±0.1870	0.3337±0.1889	—

B.2 CONSTRUCTION DETAILS

OPB construction notes. The construction is equivariant under label permutations ($Q_\theta(U_\theta P) = Q_\theta(U_\theta)P$), so no class index plays a distinguished role, and the prior encoder is identity-initialized, so the initial raw matrix $U_\theta = [e_1, \dots, e_K]$ is full column rank with a stable polar-factor backward. The prior covariance is the fixed $\tau^2 I$ with $\tau^2 = 1$ and no learned parameters. For uniform classes the anchor-energy scores of Eq. (8) are linear in z up to a class-independent term, with weights $(a/\tau^2)q_{\theta,k}$, so the readout scale is fixed by the prior alone.

EPB isometry notes. The exactness is automatic: any nonzero linear map of a scalar label is isometric up to its norm, and the normalization fixes that norm to ρ . All prior means lie on one line through the origin, so anchor norms grow with $|\tilde{y}|$: a line and a fixed-radius sphere intersect in at most two points, so isometry and equal norms cannot both hold. For $y \in \mathbb{R}^m$ ($m \leq d$) the same construction extends verbatim: take $W \in \mathbb{R}^{d \times m}$ and its thin polar factor Q_θ as in Eq. (6) with K replaced by m , and the prior $m_\theta(y) = \rho Q_\theta \tilde{y}$, isometric in the declared metric on the normalized labels, the scalar case being $m = 1$ with $Q_\theta = u$. The label metric must be declared before normalization, otherwise “isometric” has no unique meaning.

C ADDITIONAL EXPERIMENTS

C.1 MAIN RESULTS: STANDARD DEVIATIONS AND CONFIDENCE INTERVALS

Tables 7 and 8 report the same configurations as the main table with standard deviations over runs, split by column groups for width; bold and underline follow the same mean-based rule as Table 1.

Tables 9–12 report, for each setting, the paired difference of GPB over every baseline (GPB minus baseline) with a bootstrap 95% confidence interval: the differences are paired across the shared seeds, the interval is the percentile interval of $B = 10,000$ resamples of the per-run differences (fixed seed, reproducible), and the non-inferiority margin is $\delta_{\text{NI}} = 0.2$ percentage points of accuracy or $\delta_{\text{NI}} = 0.005$ in R^2 . Bold marks a lower bound above zero (significant advantage) and † a lower bound within the margin (tied); the best configuration per method is the same as in the main table.

Table 8: Test accuracy (%) and test R^2 with standard deviations over runs: CEB, DVCCA, AdaCap (regression settings only), and our method; same configurations as Table 1. GPB is separated from the references by the vertical rule.

Dataset	Backbone	CEB	DVCCA	AdaCap	GPB
MNIST	MLP	98.43±0.08	98.24±0.17	–	98.54±0.09
MNIST	CNN	99.30±0.05	99.29±0.10	–	99.32±0.05
ImageNet-100	MLP	82.77±0.22	82.99±0.43	–	84.40±0.21
ImageNet-100	CNN	82.97±0.67	83.21±0.70	–	86.08±0.21
Cora	GCN	86.27±0.38	86.90±0.94	–	88.12±0.55
IMDb	RNN	85.55±0.73	85.69±0.23	–	86.84±0.57
AG News	RNN	92.83±0.11	92.78±0.15	–	92.87±0.12
Cal. Housing	MLP	0.7894±0.0018	0.7878±0.0048	0.7758±0.0023	0.7895±0.0051
STS-B	RNN	0.2668±0.0127	0.2713±0.0218	0.1613±0.0188	0.2762±0.0152
ZINC	GCN	0.6257±0.0044	0.6294±0.0026	0.6770±0.0019	0.6344±0.0031
AgeDB	MLP	0.3317±0.0251	0.3180±0.0228	0.3482±0.0209	0.3536±0.0076
AgeDB	CNN	0.3882±0.0275	0.3205±0.1811	0.4684±0.0094	0.4700±0.0072

Table 9: Paired-difference 95% confidence intervals of GPB over the plain backbone and VIB (GPB minus baseline, per setting; classification in percentage points, regression in R^2). Bold: lower CI bound above zero; †: lower bound within the non-inferiority margin δ_{NI} (tied).

Dataset	Backbone	Base	VIB
MNIST	MLP	+0.2 [+0.2, +0.3]	+0.3 [+0.2, +0.3]
MNIST	CNN	+0.2 [+0.2, +0.2]	+0.0 [-0.0, +0.1]†
ImageNet-100	MLP	+1.3 [+1.2, +1.4]	+1.6 [+1.4, +1.9]
ImageNet-100	CNN	+2.7 [+2.5, +2.9]	+3.1 [+3.0, +3.4]
Cora	GCN	+0.4 [-0.1, +1.1]†	+1.3 [+0.7, +1.7]
IMDb	RNN	+2.0 [+1.2, +2.7]	+1.3 [+0.6, +2.3]
AG News	RNN	-0.1 [-0.2, +0.0]†	+0.2 [+0.1, +0.2]
Cal. Housing	MLP	+0.028 [+0.008, +0.048]	-0.001 [-0.005, +0.003]†
STS-B	RNN	-0.014 [-0.022, -0.006]	+0.004 [-0.017, +0.022]
ZINC	GCN	-0.000 [-0.003, +0.002]†	+0.001 [-0.004, +0.004]†
AgeDB	MLP	+0.136 [+0.104, +0.167]	+0.036 [+0.026, +0.046]
AgeDB	CNN	+0.079 [+0.066, +0.091]	+0.136 [+0.043, +0.302]
		8*/3†	8*/3†

C.2 CONTROLLED LABEL-AMBIGUITY SENSITIVITY ANALYSES

The analytic ambiguity floor is

$$\delta_{\text{noise}}(r_n) = \frac{a^2}{2\tau^2} \left[1 - (1 - r_n)^2 - \frac{r_n^2}{K - 1} \right], \quad (26)$$

which follows from Proposition A.2 with the explicit comparator of Section 3.3: its posterior mean $h_0(x) = a \sum_k \eta_k(x) q_k$ annihilates the bias term, its covariance $\tau^2 I$ annihilates V_0 , and $\eta_k(x)$ equals $1 - r_n$ on the true class and $r_n/(K - 1)$ otherwise, so $1 - \|\eta\|_2^2 = 1 - (1 - r_n)^2 - r_n^2/(K - 1)$. It grows from 0 to 11.2 nats as r_n runs over the noise grid.

Training evaluates the population objective by weighting all K^2 clean/observed-label pairs by the noise model, with a descending- β warm start to remove avoidable optimization error at the small- β end; the comparator risk $C_{\text{comp}}(r_n)$ is estimated by Monte Carlo. The population-weighted experiment of Section 3.3 uses a fixed frame and the standard trainable posterior variance. Figure 5 changes these choices one at a time. Fixing the posterior variance to $\sigma^2 \equiv \tau^2$ leaves the measured mismatch on the analytic floors, confirming that the result is not an artifact of posterior-covariance flexibility. Replacing the fixed frame by the trainable polar frame gives the same result, so the floor is invariant to the learned rigid frame rotation.

The third panel replaces the exact population weights by sampled noisy labels. It shows the expected finite-sample displacement from the population floor at weak compression and convergence toward the empirical floor as compression strengthens. Six sampled-label configurations at large β fail the

Table 10: Paired-difference 95% confidence intervals of GPB over SVIB and NIB (GPB minus baseline, per setting; classification in percentage points, regression in R^2). Bold/ $\dagger$ as in the baselines table.

Dataset	Backbone	SVIB	NIB
MNIST	MLP	+0.4 [+0.3 , +0.5]	+0.3 [+0.2 , +0.3]
MNIST	CNN	+0.1 [+0.1 , +0.1]	+0.1 [+0.1 , +0.2]
ImageNet-100	MLP	+1.4 [+1.1 , +1.7]	+2.1 [+1.8 , +2.2]
ImageNet-100	CNN	+3.0 [+2.4 , +3.9]	+3.2 [+2.9 , +3.5]
Cora	GCN	+1.7 [+0.2 , +3.4]	+1.8 [+1.1 , +2.4]
IMDb	RNN	+0.8 [+0.2 , +1.4]	+0.7 [+0.2 , +1.2]
AG News	RNN	+0.0 [-0.1, +0.1] $\dagger$	-0.0 [-0.2, +0.1] $\dagger$
Cal. Housing	MLP	-0.000 [-0.006, +0.004]	+0.003 [-0.001, +0.009] $\dagger$
STS-B	RNN	+0.006 [-0.021, +0.028]	+0.035 [+0.011 , +0.058]
ZINC	GCN	+0.002 [-0.001, +0.004] $\dagger$	+0.017 [+0.010 , +0.026]
AgeDB	MLP	+0.015 [-0.010, +0.041]	+0.083 [+0.062 , +0.100]
AgeDB	CNN	+0.138 [+0.045 , +0.303]	+0.136 [+0.040 , +0.302]
		7*/2 $\dagger$	10*/2 $\dagger$

Table 11: Paired-difference 95% confidence intervals of GPB over CEB and DVCCA (GPB minus baseline, per setting; classification in percentage points, regression in R^2). Bold/ $\dagger$ as in the baselines table.

Dataset	Backbone	CEB	DVCCA
MNIST	MLP	+0.1 [+0.0 , +0.2]	+0.3 [+0.2 , +0.4]
MNIST	CNN	+0.0 [-0.0, +0.1] $\dagger$	+0.0 [-0.1, +0.1] $\dagger$
ImageNet-100	MLP	+1.6 [+1.5 , +1.7]	+1.4 [+1.1 , +1.7]
ImageNet-100	CNN	+3.1 [+2.4 , +3.7]	+2.9 [+2.4 , +3.5]
Cora	GCN	+1.8 [+1.5 , +2.3]	+1.2 [+0.4 , +1.9]
IMDb	RNN	+1.3 [+0.3 , +2.1]	+1.1 [+0.8 , +1.5]
AG News	RNN	+0.0 [-0.1, +0.2] $\dagger$	+0.1 [-0.0, +0.2] $\dagger$
Cal. Housing	MLP	+0.000 [-0.003, +0.004] $\dagger$	+0.002 [-0.005, +0.009]
STS-B	RNN	+0.009 [-0.009, +0.025]	+0.005 [-0.020, +0.026]
ZINC	GCN	+0.009 [+0.005 , +0.011]	+0.005 [+0.002 , +0.007]
AgeDB	MLP	+0.022 [+0.006 , +0.038]	+0.036 [+0.012 , +0.057]
AgeDB	CNN	+0.082 [+0.064 , +0.099]	+0.149 [+0.065 , +0.310]
		8*/3 $\dagger$	8*/2 $\dagger$

explicit comparator-objective gate (twelve stored rows because both best and final checkpoints are retained); they are marked as optimization failures rather than interpreted as violations of Eq. (12). The population-weighted main grid and both structural controls satisfy the gate at every point.

C.3 ALIGNMENT AND SEPARATION: DEFINITIONS AND DIAGNOSTICS

This section collects the diagnostic panels deferred from Section 3.4: the decoupling scatter, the separation-bound and slack readings, and the regression diagnostics, together with the finite-bin definitions. All statistics are first computed within each seed and then aggregated across seeds; class or bin pairs are not treated as independent replicates. The per-class Jensen check and the per-pair separation bound are verified numerically on every scale, β , seed, and class (no violation exceeds 10^{-6}). Every checkpoint in both requested grids is present, and no collapse or variance-explosion point is removed.

The deferred classification panels complete the picture (Fig. 6). The decoupling scatter (a) shows MC-10 accuracy staying within 82–86% while $\hat{D}$ spans three orders of magnitude: mechanism and task metric are decoupled. Panel (b) shows the raw bound rising toward the declared $\sqrt{2}a$ exactly where the mismatch corrections shrink, so the slack narrows where the bound is informative; CEB’s mismatch is plotted descriptively, without a bound or slack entries, since its reference has no declared Δ . The slack percentiles (c) stay non-negative at every reported point: the bound is consistent but conservative.

Table 12: Paired-difference 95% confidence intervals of GPB over the external non-IB baselines NCBD (classification settings only) and AdaCap (regression settings only); GPB minus baseline, classification in percentage points, regression in R^2 . Bold/† as in the baselines table.

Dataset	Backbone	NCBD	AdaCap
MNIST	MLP	+0.6 [+0.4, +0.8]	—
MNIST	CNN	+0.1 [+0.1, +0.2]	—
ImageNet-100	MLP	+0.8 [+0.6, +0.9]	—
ImageNet-100	CNN	+5.5 [+4.3, +6.9]	—
Cora	GCN	+0.8 [+0.0, +1.9]	—
IMDb	RNN	+0.2 [-0.5, +0.7]	—
AG News	RNN	+0.5 [+0.2, +0.8]	—
Cal. Housing	MLP	—	+0.014 [+0.009, +0.019]
STS-B	RNN	—	+0.115 [+0.086, +0.135]
ZINC	GCN	—	-0.043 [-0.046, -0.041]
AgeDB	MLP	—	+0.005 [-0.007, +0.021]
AgeDB	CNN	—	+0.002 [-0.004, +0.006] [†]
		6*/0 [†]	2*/1 [†]

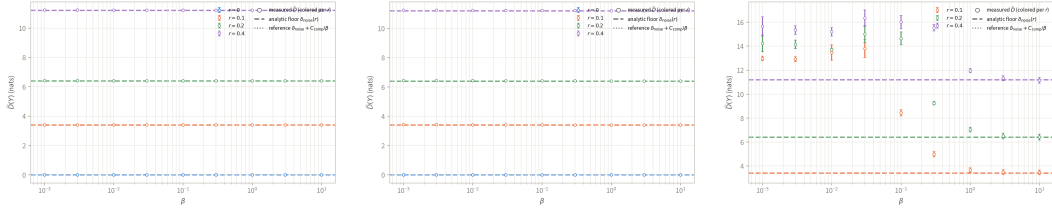


Figure 5: Sensitivity analyses for the controlled label-ambiguity experiment. (a) Posterior variance fixed to the prior variance. (b) Trainable polar frame. (c) Labels sampled rather than integrated with their exact population weights; failed comparator-gate configurations are marked. Dashed lines denote the analytic population ambiguity floors. Points are means and error bars are standard deviations over seeds.

Regression. We partition the normalized test labels into 20 equal-width bins and merge any bin containing fewer than 30 examples. Let $\bar{y}_b$ be the mean label of bin B_b , $\hat{\mu}_b$ its posterior center, $\hat{D}_b$ its average sample-to-own-anchor mismatch, and $h_b := \text{diam}(B_b)$. The finite-bin lower bound is

$$\widehat{\text{LB}}_{bc} = \rho|\bar{y}_b - \bar{y}_c| - \sqrt{2\tau^2\hat{D}_b} - \sqrt{2\tau^2\hat{D}_c} - \rho(h_b + h_c), \quad (27)$$

where the last term is the quantization correction of Remark A.1. On Housing with $\rho = 6$, mismatch falls from 0.264 to 0.026, the axial slope remains 0.73–0.77, and the mismatch is almost entirely axial (about 0.2 against 0.004 off-axis), while R^2 changes only from 0.786 to 0.767 over the sweep: a tenfold mismatch reduction need not imply a comparable task-metric gain. Figure 6d plots the actual bin-center distances against the declared isometric distances at $\rho = 6$: the cloud hugs a line of slope 0.73–0.77 below the identity line at all four β values, so the posterior follows the declared metric structure at about three quarters of its scale, and the scale-transfer ratio is essentially β -invariant. After the quantization correction, 43% of bin pairs retain a positive bound. Figure 7 shows the two regression sweep panels: the mismatch- β curves with their axial/off-axis decomposition (the dotted off-axis contribution stays below 0.01 nats, so the posterior lies essentially on the declared axis), and the R^2 -against- $\hat{D}$ scatter whose band stays within 0.75–0.79 while $\hat{D}$ spans orders of magnitude.

Normalization and variance extremes. The empirical mismatch normalizes by the declared prior variance $\tau^2 = 1$, whereas the exact Gaussian KL uses every sample’s and coordinate’s own posterior variance $\sigma_\phi^2(x)$. They therefore need not coincide: averaged over the ImageNet-100 grid, $\hat{D} = 41.2$, the KL mean term is 46.5, and the KL variance term is 51.4. On Housing, a small number of weak-compression, large- ρ posterior variance explosions raise the across-grid mean KL variance term to 2.8×10^6 ; these checkpoints remain in all plots. The unnormalized axial/off-axis components and total KL are therefore reported alongside $\hat{D}$.

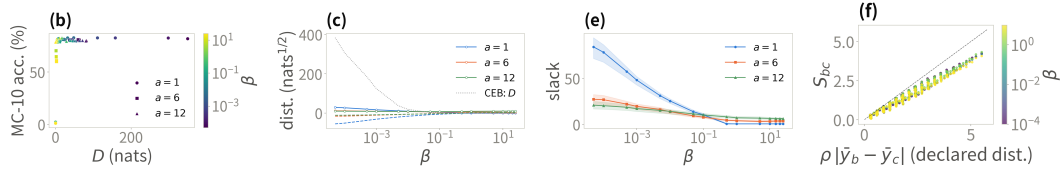


Figure 6: Alignment and separation diagnostics deferred from the main text. (a) MC-10 accuracy against $\hat{D}$ (color = β , marker = a). (b) Mean separation (solid) and raw lower bound (dashed) against β ; gray dotted: CEB, descriptive. (c) Slack percentiles. (d) Cal. Housing bin-pair separation against the declared prior distance at $\rho = 6$ (color = β ; identity line: exact transfer).

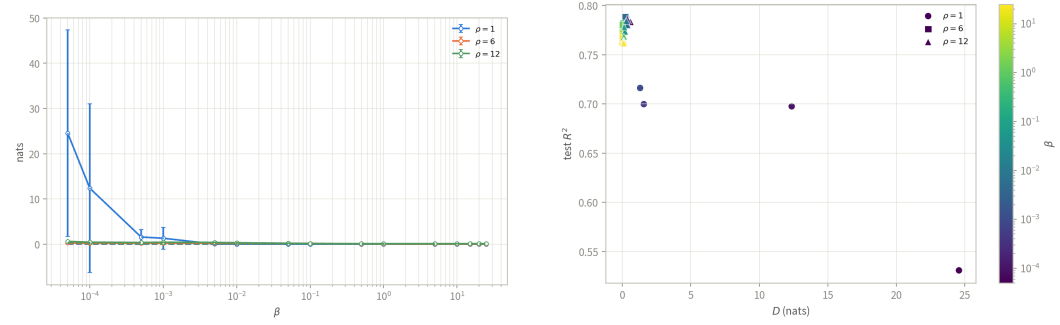


Figure 7: Cal. Housing mismatch sweep panels (deferred from the main text). (a) Anchor mismatch $\hat{D}$ against β , by isometric scale ρ , decomposed into axial and off-axis contributions. (b) Test R^2 against $\hat{D}$; color denotes β and marker shape ρ .

C.4 ADVERSARIAL ROBUSTNESS DETAILS

Targeted PGD uses 20 steps, EOT-10 gradients, 10-sample MC predictions, and budgets $\varepsilon_\infty = 0.3$ and $\varepsilon_2 = 6$. Curves are shown only when clean stochastic accuracy exceeds 90%. At maximum compression, OPB ($a = 6, 12$) reaches 31.1%/30.5% targeted success under L_∞ , while transfer attacks from CEB to OPB reach at most 2.0%/3.4% under L_∞/L_2 , far below OPB’s white-box attack. The transfer result is a gradient-masking check rather than a causal geometry test.

C.5 FACTOR-WISE ABLATIONS

The factor-wise ablation ladder switches geometry, anchor scale, readout, and the IB term separately; each variant is evaluated at its best validation-selected configuration (Table 13). The complete means are reported in Table 13. The paired bootstrap intervals are in the main text (Table 3) for the classification contrasts and in Table 14 for the regression mirrors; all paired differences use the shared seeds at each variant’s best validation-selected configuration, and the intervals are 95% bootstrap percentile CIs with $B = 10,000$ and a fixed seed. The factor ranking and the sign-flip and negative-result patterns are read in the main text (Section 3.6).

NCM-learn removes the bottleneck entirely: the encoder output is used deterministically ($z = \mu_\phi(x)$, no posterior and no KL) and classified by an energy readout on trainable prototypes. *NCM-ortho* is the same, except that the prototypes are fixed to the orthogonal frame $a e_k$ (fixed geometry, still no IB). *CEB-energy* and *CEB-tied* keep CEB’s trainable free reference and KL but replace its free head by the energy readout (resp. the tied projection). *OPB-FS* and *EPB-FS* keep the orthogonal frame and the energy (resp. tied) readout but make the anchor radius learnable, so only the fixed scale is switched off. *GPB-L* is GPB’s free-head counterpart on the classification settings (identical orthogonal-frame prior, free linear head), and *EPB-L* and *EPB* are the regression counterparts of GPB-L and GPB on Housing, keeping the isometric-axis prior with a free head (resp. the tied projection). *OPB-RV* freezes all trainable prior parameters on the classification settings, and *EPB-RV* freezes all trainable prior parameters on the regression settings.

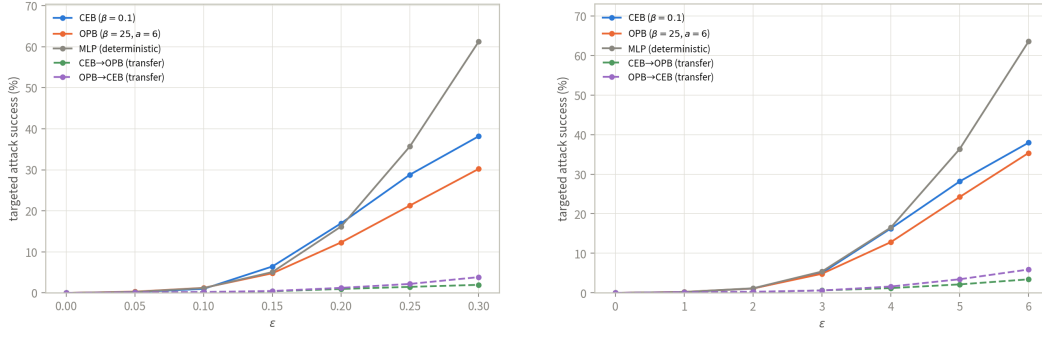


Figure 8: Transfer PGD curves on MNIST (gradient-masking check). Targeted adversaries are generated on one model and evaluated on another; CEB-to-OPB transfer remains far below OPB’s own white-box attack. The targeted white-box curves are in the main text.

Table 13: Attribution ablation: best-configuration test metrics (mean $\pm$ std over seeds). Configurations are selected by the mean validation metric (accuracy on MNIST and ImageNet-100, R^2 on Housing), applied identically to every variant, so all comparisons remain paired by seed and the test split is never touched by selection. One OPB-FS configuration (ImageNet-100, $\beta = 10^{-4}$, $a = 1$) diverged during training (free scales grow without penalty at negligible β) and is excluded.

variant	MNIST (Acc %)	ImageNet-100 (Acc %)	Housing (R^2)
Base	98.3 $\pm$ 0.1	83.1 $\pm$ 0.1	0.762 $\pm$ 0.024
NCM-learn	98.2 $\pm$ 0.2	82.2 $\pm$ 0.4	—
NCM-ortho	98.1 $\pm$ 0.1 ($a = 1$)	82.3 $\pm$ 0.6 ($a = 1$)	—
CEB	98.4 $\pm$ 0.1 ($\beta = 0.01$)	82.8 $\pm$ 0.2 ($\beta = 0.01$)	0.789 $\pm$ 0.002 ($\beta = 0.0001$)
CEB-energy	98.2 $\pm$ 0.1 ($\beta = 0.001$)	81.6 $\pm$ 0.2 ($\beta = 0.001$)	—
GPB-L	98.4 $\pm$ 0.1 ($\beta = 0.1$, $a = 12$)	83.4 $\pm$ 0.7 ($\beta = 1$, $a = 12$)	—
GPB	98.4 $\pm$ 0.1 ($\beta = 0.1$, $a = 12$)	84.2 $\pm$ 0.1 ($\beta = 1$, $a = 12$)	—
OPB-RV	98.3 $\pm$ 0.0 ($\beta = 1$, $a = 6$)	83.4 $\pm$ 0.5 ($\beta = 10$, $a = 12$)	—
OPB-FS	98.5 $\pm$ 0.0 ($\beta = 0.1$, $a = 12$)	84.3 $\pm$ 0.2 ($\beta = 1$, $a = 12$)	—
CEB-tied	—	—	0.781 $\pm$ 0.006 ($\beta = 0.01$)
EPB-L	—	—	0.791 $\pm$ 0.003 ($\beta = 0.0001$, $a = 6$)
EPB	—	—	0.788 $\pm$ 0.004 ($\beta = 0.0001$, $a = 6$)
EPB-RV	—	—	0.789 $\pm$ 0.002 ($\beta = 0.001$, $a = 6$)
EPB-FS	—	—	0.787 $\pm$ 0.008 ($\beta = 0.001$, $a = 6$)

Reading the regression mirrors. The frame-trainability contrast is a null on Housing (Table 14): freezing all trainable prior parameters costs nothing, so the trainable rotation carries no benefit in regression, unlike the classification settings where it adds +0.83 on ImageNet-100 (Table 3).

C.6 REGRESSION COMPRESSION CURVES

The main text shows the classification compression sweep. This companion figure gives the complete California Housing regression sweep under the same β grid, including the weak-compression variance explosion and the strong-compression EPB plateau. The curves are diagnostics of achieved KL and task performance, not tests of the alignment rate.

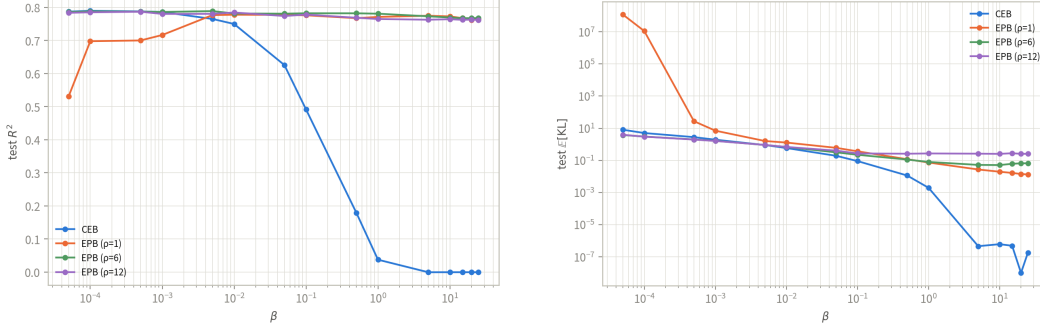
C.7 ANCHOR-SCALE SENSITIVITY

The main comparison selects the prior scale on the five-value grid $\{1, 3, 6, 9, 12\}$ (Sec. 3.1); Fig. 10 reads the full five-point scale grid at weak, medium, and strong compression on the two representative settings, complementing the β -axis sweeps of Sec. 3.7 with the scale-axis view.

On both settings the scale matters only once compression bites: the weak-compression curves are flat (within 3.4 points on ImageNet-100), and at $\beta \geq 1$ scales below 4 collapse (to chance accuracy on ImageNet-100, to zero R^2 on AgeDB) while every scale above 6 stays on a plateau; the operating point is therefore sensitive to the scale only through this collapse boundary, and the selected scale lies inside the retained band.

Table 14: Regression mirrors of the factor-wise paired differences (R^2 on Housing; 95% bootstrap percentile CIs; positive values favor the first term).

factor	comparison	Housing
readout (EPB)	EPB — EPB-L	-0.00 [-0.01, -0.00]
geometry (EPB)	EPB-FS — CEB-tied	+0.01 [-0.00, 0.01]
fixed scale (EPB)	EPB — EPB-FS	+0.00 [-0.01, 0.01]
frame trainability (EPB)	EPB — EPB-RV	-0.00 [-0.00, 0.00]

Figure 9: California Housing compression sweep (MLP). Left: test R^2 ; right: realized conditional KL over β . EPB maintains a task plateau through strong compression, while the $\rho = 1$ weak-compression points exhibit a large variance term.

C.8 INFORMATION-PLANE ESTIMATES

Estimator protocol. $I(X; Z)$ is estimated by the InfoNCE lower bound (van den Oord et al., 2018) ($\log N_{\text{neg}} - \mathcal{L}_{\text{NCE}}$, $N_{\text{neg}} = 4096$; one critic per run, trained on the training split, the bound read off the test split with $z \sim q(z|x)$). $I(Y; Z)$ is read on the same sampled posterior: $H(Y) - \text{CE}$ with the cross-entropy averaged over $M = 20$ draws $z \sim q(z|x)$ per test point (still a lower bound, as $\text{CE} \geq H(Y|Z)$). The residual $I(X; Z|Y)$ is estimated directly rather than as a difference: every $q(z|x)$ is a known Gaussian, so the class-conditional mixture $q(z|y) = \frac{1}{N_y} \sum_{j: y_j=y} q(z|x_j)$ over the test samples is evaluated exactly by logsumexp, and $I(X; Z|Y) = \mathbb{E}_{x,y} \text{KL}(q(z|x) \| q(z|y))$ is read by Monte Carlo—an exact estimate up to z -sampling noise and the finite per-class test sample, not a difference of bounds. The certificate-plane axis is drawn linear because the collapsed CEB trajectory lands on zero. The MNIST panels appear in Fig. 3(c,d) of the main text.

Matched-compression performance. Table 15 pairs each CEB point with the OPB configuration of nearest estimated $I(X; Z)$. At ≈ 2.0 nat, CEB ($\beta = 0.5$) retains 86.0% while OPB ($a = 1, \beta = 0.5$) retains 97.8%; at ≈ 2.5 nat ($\beta = 0.1$) the two tie (98.2% vs. 98.1%). The directly estimated residual is comparable or slightly larger for OPB at both points (0.41 vs. 0.47 nat at $\beta = 0.1$; 0.20 vs. 0.21 nat at $\beta = 0.5$), so OPB’s task plateau is not bought by extra residual compression. These are paired empirical comparisons at two estimated compression levels, not a causal attribution to geometry.

C.9 INFORMATION-PLANE PANELS FOR CAL. HOUSING

Figure 11 completes panels (c)–(d) of Fig. 3 in the main text with the regression setting, under the same β grid and five runs per point; here $I(Y; Z)$ is the Gaussian approximation $\frac{1}{2} \ln(\text{Var}(y)/\text{Var}(y - \hat{y}_{\text{MC}}))$ read on the sampled posterior, and the residual $I(X; Z|Y)$ is estimated directly rather than as a difference of the two: $\hat{q}(z|y)$ is the Gaussian-kernel-weighted mixture $\sum_j w_j(y) q(z|x_j)$ over the test samples (w_j a softmax in $-(y - y_j)^2/2h^2$, truncated to the 2048 nearest labels and renormalized, with Silverman’s h fitted on the training split), the mixture density is evaluated exactly by logsumexp, and $I(X; Z|Y) = \mathbb{E}_{x,y} \text{KL}(q(z|x) \| \hat{q}(z|y))$ is read by Monte Carlo; every term is a KL between proper densities, so the estimate is non-negative by construction (doubling h shifts it by a median 6.5%, at most 12%). The certificate panel now reads directly: CEB’s residual falls from ≈ 2.9 nat at weak compression to near zero together with its task information, whereas EPB at $\rho = 6, 12$ ends strong compression on a small positive residual plateau of 0.05–0.15 nat with $I(Y; Z)$ held at 0.52–0.67 nat—compression again stops near the

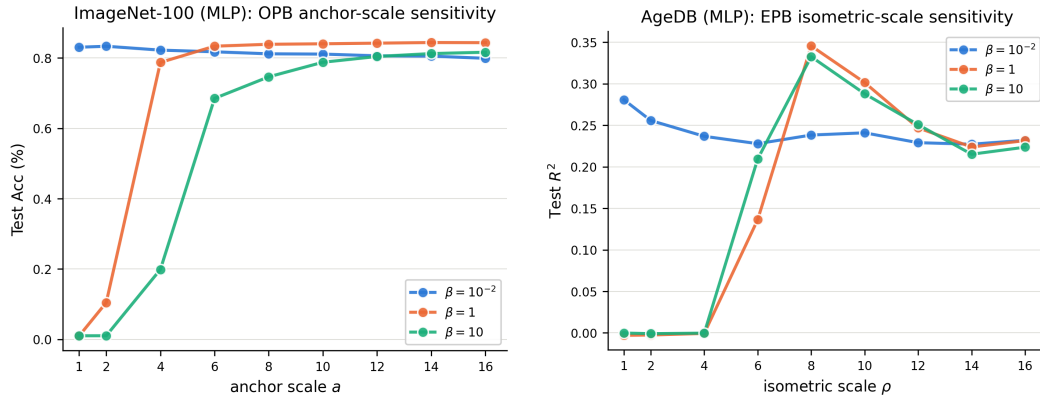


Figure 10: Anchor-scale sensitivity: (a) ImageNet-100 (MLP) OPB test accuracy against the anchor scale a ; (b) AgeDB (MLP) EPB test R^2 against the isometric scale ρ ; $\beta \in \{10^{-2}, 1, 10\}$ (weak/medium/strong compression).

Table 15: Matched-compression information-plane comparison (means; standard deviations are in the stored evaluation report). The OPB configuration is the one whose estimated $I(X; Z)$ is nearest to the CEB point.

Setting	CEB			config.	OPB		
	$I(X; Z)$	$I(X; Z Y)$	Acc.		$I(X; Z)$	$I(X; Z Y)$	Acc.
MNIST, $\beta = 0.1$	2.54	0.41	98.2	$a = 1, \beta = 0.1$	2.55	0.47	98.1
MNIST, $\beta = 0.5$	1.96	0.20	86.0	$a = 1, \beta = 0.5$	2.01	0.21	97.8

task-information level, now on a non-negative direct estimate. The information plane is informative: CEB compresses itself off the task ($R^2 = 0.49$ at $\beta = 0.1$, 0.18 at $\beta = 0.5$) while every EPB curve keeps $R^2 \geq 0.76$ flat out to $\beta = 25$.

D RELATED WORK

Information bottleneck and conditional compression. The information bottleneck (IB) principle formulates representation learning as retaining information about a target while discarding information about the input (Tishby et al., 2000). Deep VIB turns this principle into a neural objective by using a reparameterized stochastic encoder and a variational marginal prior (Alemi et al., 2017). Nonlinear and non-parametric bottleneck variants study alternative variational bounds and distance-based estimates of the information in a representation (Kolchinsky et al., 2019). The deterministic-scenario analysis of Kolchinsky & Tracey (2019) further shows that the usual IB multiplier can fail to sweep an interior information-accuracy trade-off when labels are deterministic; the squared-compression baseline in our experiments follows this line of work.

The conditional entropy bottleneck (CEB) replaces the marginal reference of VIB by a label-conditional reference and thereby targets conditional residual information (Fischer, 2020). Comparisons of variational bounds clarify when this conditional reference is tighter than a marginal one (Geiger & Fischer, 2020). CEB has also been applied to large-scale visual representations and robustness evaluations (Fischer & Alemi, 2020; Lee et al., 2021), while multivariate extensions retain the same variational bottleneck viewpoint (Abdelaleem et al., 2025). These methods leave the relationships among distinct conditional reference means unspecified. GPB keeps the CEB posterior and its single conditional KL, but restricts the reference family itself: OPB declares a class frame and EPB declares a label-distance-preserving axis.

Information bottleneck for robust representations. Several works use a bottleneck to suppress nuisance or non-robust information. The HSIC bottleneck adds layer-wise dependence penalties and analyzes adversarial robustness (Wang et al., 2021). Adversarial Information Bottleneck explicitly incorporates adversarial objectives (Zhai & Zhang, 2022), and an NLP-specific variant separates

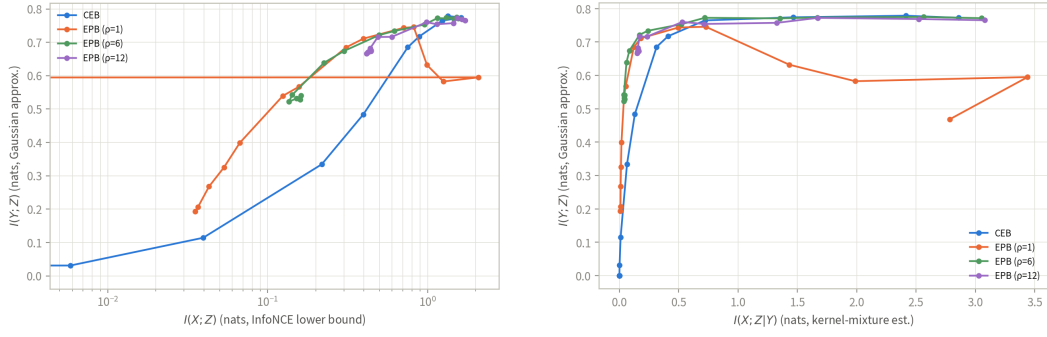


Figure 11: Information plane (a) and certificate plane (b) for Cal. Housing, companion panels to Fig. 3(c,d): identical estimators, β grid (log x -scale in both panels), lines connect β in ascending order. CEB one curve per panel; EPB three ($\rho \in \{1, 6, 12\}$).

task-relevant robust features from manipulable features (Zhang et al., 2022). Causal Information Bottleneck uses a causal decomposition for the same robust-feature goal (Hua et al., 2022). These approaches modify the information criterion, feature selection, or training examples. In contrast, the geometric constraint in GPB is placed on the conditional prior and is evaluated with the ordinary CEB KL; the robustness experiments therefore test whether a declared prior geometry is transferred to the posterior rather than introducing adversarial training as an additional objective.

Geometric classification representations. Neural-collapse studies document an approximately simplex-ETF arrangement of class means and classifier directions near the terminal phase of training (Papayan et al., 2020; Zhu et al., 2021). The connection between neural collapse, supervised contrastive learning, and an information bottleneck has also been analyzed explicitly (Wang & Palmer, 2023). Other works impose related geometry directly: orthogonal classifier weights can improve adversarial robustness (Xu et al., 2022); ETF classifiers have been used for early exit (Ji et al., 2023); and fixed or hierarchy-aware frames have been used to reduce classifier bias or mistake severity (Tong & Davis, 2023). Neural Collapse by Design trains features and prototypes jointly on the unit hypersphere so that the global optimum of the NONL contrastive objective is exactly the simplex ETF (Koromilas et al., 2026); we include it as a classification-only comparison baseline. Equidistant prototype embeddings provide an earlier prototype-based classification example (Deng et al., 2014).

These works constrain features, classifier weights, or prototypes through a classification loss or a fixed readout. OPB addresses a different object: the label-conditional Gaussian prior in the bottleneck. Its polar/QR frame is recomputed from the prior encoder, and the anchor-energy classifier reads the same frame without introducing an independent classifier scale. Thus the geometric class code and the conditional KL are coupled during training.

Table 16 summarizes this object-level distinction. The geometric primitive is shared with several preceding lines of work, but its role in the objective is different: OPB anchors are constrained class centers that serve simultaneously as Gaussian-prior means, KL targets, and readout anchors. In the shared-covariance GPB instance, the induced readout is an orthogonal classifier as a consequence of the Gaussian Bayes rule. It is not an ETF construction: OPB has Gram matrix $a^2 I_K$ (zero pairwise inner products), whereas a centered simplex ETF has constant negative off-diagonal inner products in a $(K-1)$ -dimensional subspace. The frame is recomputed from the prior encoder and may rotate, rather than fixing a coordinate frame or optimizing a separate prototype table.

Distance-preserving regression and geometric priors. Isometric representation learning has traditionally focused on preserving distances on a data manifold, including deep extensions of isomap-style mappings (Pai et al., 2022). Orthogonal label regression has also been used as a discriminative device in domain adaptation (Luo et al., 2023). These methods motivate using a declared label metric, but they generally constrain a deterministic feature map rather than the mean map of a conditional Gaussian information-bottleneck prior. EPB uses the scalar label case: after standardization, prior means lie on a normalized axis with a fixed scale, so latent prior distances equal the declared label distances up to that scale. The same axis is also the tied regression readout and the target of

Table 16: Conceptual boundary between geometric representation methods and GPB. The entries describe the object constrained by the method, not whether a particular implementation may contain additional components. GPB reuses familiar geometric primitives while placing them in the conditional prior used by the bottleneck and using the same prior geometry for prediction.

Approach	Constrained object	Enters an IB KL?	Readout tied to geometry?	Stochastic posterior?
Prototype methods (Deng et al., 2014)	prototypes or features	No	Usually no/partial	Usually no
ETF / orthogonal classifiers (Xu et al., 2022; Ji et al., 2023)	classifier weights or feature directions	No	Yes	Usually no
Fixed or hierarchy-aware frames (Tong & Davis, 2023)	feature coordinates or a frame	No	Method-dependent	Usually no
CEB (Fischer, 2020)	label-conditional reference	Yes (conditional KL)	No (free head)	Yes
GPB (OPB / EPB)	label-conditional Gaussian prior means with shared covariance	Yes (conditional KL)	Yes (tied rule)	Yes

the conditional KL. Thus EPB is a conditional-prior use of distance geometry, not a new distance-preserving primitive. The regression information-bottleneck literature, including divergence-based bounds, provides complementary compression objectives rather than this geometric prior construction (Yu et al., 2024). AdaCap pairs a permutation-based contrastive loss with a Tikhonov closed-form output layer for small-sample tabular regression (Belucci et al., 2026); we include it as a regression-only comparison baseline.

Geometry-aware information bottlenecks. Recent work has used the phrase geometric information bottleneck in several distinct settings. A variational recommender learns a geometric prior for explanation generation (Yan et al., 2024); a data-scarcity formulation penalizes latent curvature and intrinsic dimension (Katende, 2025); and activation steering uses orthogonal subspaces to control language-model interventions (Doan et al., 2026). A concurrent analysis studies the relationship between information compression and class-wise geometric compression in CEB networks (Adilova et al., 2026). These studies make geometry an important representation concern, but they do not use the same conditional Gaussian prior for both compression and prediction, nor do they provide the OPB/EPB alignment-to-separation construction considered here.